Discrete Math Class Notes

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0 Introduction to Graph Theory

Most of this can be found early in Chapter 3 (Set stuff) or 11 (Graph Stuff) of the text.

0.1 Basic Notation

We start with an imprecise idea of a set: 'a collection of distinct elements'. Following convention, we will usually denote sets by capital letters such as A and X and elements as small letters, such as a, b and z. We write $a \in X$ to denote that 'a is and element of the set X', or simply that 'a is in X'.

Normally our elements will be numbers or variables, but we will also consider sets whose elements are other mathematical constructions, such as sets. We usually use some nice calligraphic letter such as $\mathcal{S}$ for a set of sets.

Note

This is where trouble arises in set theory, when you want to allow sets that contain themself. Russel defined the set $\mathcal{S}$ of all sets that contain themself and asked if $\mathcal{S}$ is in $\mathcal{S}$. It was a nasty trick that caused a lot of good people a lot of bad times. We won't do this, our sets never contain themself. This allows us to ignore most of the exciting mess of formal set theory.

Further, we consider only finite sets, or if infinite, countably infinite sets. This allows us to ignore more of the mess of set theory.

Some typical sets we will meet are $\{1, 2, 3\}$, $[n] = \{1, 2, \ldots, n\}$, $\mathbb{Z}$, $\{a, b, \ldots, f\}$ $\{\{1, 2\}, \{2, 3\}, \{1, 4, 5\}\}$, $\{(1, 2), (2, 1), (1, 4)\}$, $\emptyset = \{\ \}$.

Given two sets A and B, A *is a subset of* B, written $A \subseteq B$, if every element of A is an element of B.

Problem 0.1. Given this definition, is $A \subseteq A$. Where $\emptyset = \{\ \}$ is *the empty set*, is $\emptyset \subseteq A$? Is $\{3, 1, 2, 2\} \subseteq \{1, 2, 3\}$.

We will build sets from others using set-builder notation. Given a set U (called a *universe*) and a statement $p(x)$ about elements of U,

$$\{x \in U \mid p(x)\}$$

is the set of elements in U for which p is true. Often the universe is understood from the context and we omit it, and simply write $\{x \mid p(x)\}$.

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For example the set $2\mathbb{N} = \{0, 2, 4, 6, \dots\}$ can be described by

$$2\mathbb{N} = \{x \in \mathbb{Z} \mid x \geq 0, 2 \text{ divides } x\},$$

or as the universe is is easily infered $2\mathbb{N} = \{x \mid x \geq 0, 2 \text{ divides } x\}$. One can even use a function of x:

$$2\mathbb{N} = \{2x \mid x \geq 0\}.$$

We will also consider a couple variations on the concept of a set: *multisets* and *ordered sets*. The sets $\{a, b\}$, $\{b, a\}$, and $\{a, a, b\}$ are the same as sets.

When we want to distinguish the order of the element, we use the ordered set (a, b) or (b, a). These are different.

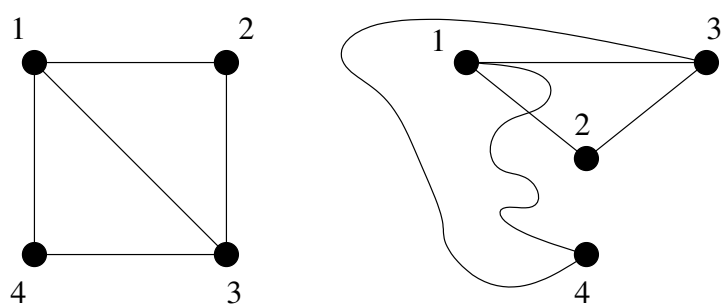
0.2 Basic Graph Definitions

Definition 0.1. A graph $G = (V, E)$ consists of a non-empty set V of *vertices* and a set E of 2-element subsets of V, called *edges*.

For example

$$G = (\{1, 2, 3, 4\}, \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}, \{1, 3\}\})$$

is a graph. It has four vertices and five edges. As this mess of brackets gets troublesome to write, we drop them when it doesn't confuse things an write an edge $\{x, y\}$ as xy. The graph G can be represented pictorially as



We often write $V(G)$ for V and $E(G)$ for E if we don't define the these sets for a graph G explicitly.

There are a lot of definitions related to the fact that $e = uv$ is an edge of G. We say

- the vertices u and v are *adjacent*, written $u \sim v$,

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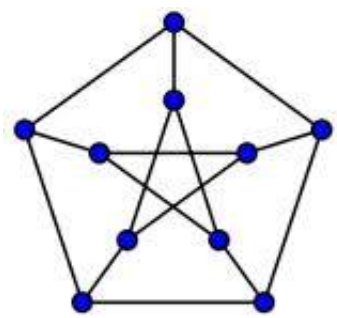
- u and v are *neighbours*,
- u and v are the *endpoints* of e,
- u (and v) is incident with e,

Here is a graph, called the *Petersen Graph*, and denoted P, that some people like to put on t-shirts, and refer to fondly as the *universal counter-example*:



Note

Recall the always useful *factorial* notation:

$$n! = n \times (n-1) \times \cdots \times 1.$$

Problem 0.2. How many graphs are there with the vertex set $\{1, 2, 3, 4\}$.

Note

One can generalise the idea of a graph to

- *digraphs* by making edges ordered sets: uv and vu are different,
- *multigraphs* by making the edgeset E a multiset,
- *hypergraphs* by allowing edges to contain more than two vertices,
- *non-simple graphs* by allowing edges of the form uu.

We mostly don't consider these generalisations, but maybe will use digraphs a bit.

Isomorphisms

Often we don't care about the names of the vertices, and want to consider graphs to be the same if they look the same when ignoring these vertex names. For this we make the following defintion.

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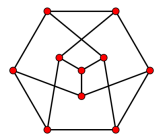
Definition 0.2. An *isomorphism* is a bijective map $f : V(G) \to V(H)$ such that for $u, v \in V(G)$,

$$uv \in E(G) \iff f(u)f(v) \in E(H).$$

If there is an isomorphism f from G to H, we say G and H are *isomorphic*, and write $G \cong H$.

Problem 0.3. Is  isomorphic to the Petersen graph?

Theorem 0.3. *Isomorphism is an equivalence relation. That is:*

i. $G \cong G$ *(reflexivity)*

ii. $G \cong H \Rightarrow H \cong G$ *(symmetry)*

iii. $G \cong H$ *and* $H \cong J \Rightarrow G \cong J$ *(transitivity)*

We prove part ii of this theorem.

Proof. Assume that $G \cong H$, that is, that there is an isomorphism $f : G \to H$. To show $H \cong G$, we show that $f^{-1} : H \to G$ is an isomorphism. As f is a bijection, so is f^{-1}, so what we have to show is that for $u, v \in V(H)$

$$uv \in E(H) \iff f^{-1}(u)f^{-1}(v) \in E(G).$$

As $f : G \to H$ is an isomorphism, we have that for any $u', v' \in V(G)$,

$$f(u')f(v') \in E(H) \iff u'v' \in E(G).$$

Now let u and v be in $V(H)$. As f is a bijection, there are u' and v' in $V(G)$ such that $f(u') = u$ and $f(v') = v$; that is to say, that $f^{-1}(u) = u'$ and $f^{-1}(v) = v'$. Plugging these equalities into the second equation above, we get the first, as needed. $\square$

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Note

'Assume/Let': We were proving an implication: 'A implies B'. The most direct way to do this is to *assume* A and then *show* B. When writing a proof, it is good practice to write your assumptions explicitly. Alternately we could have said 'Let $G \cong H$'. I went with 'assume that' instead because we try not to let the main verb of a sentence be written symbolically. To use 'Let' I would have written 'Let G be isomorphic to H.'
'what we have to show': Our goal is to show $G \cong H \Rightarrow H \cong G$. There is a lot of definition involved in this. The first two sentences of the proof reduces this to the heart of the problem: showing the $u, v \in V(H)$ implies $uv \in E(H) \iff f^{-1}(u)f^{-1}(v) \in E(G)$. Phrases like 'what we have to show' or 'what remains to be shown' tell the reader that they can forget what came before, and focus on the new goal. We finished the proof with 'which is what we had to show.' This refers back to the the phrase 'what we have to show'.
Prove, show, demonstrate: These words are used interchangably. They all mean prove. Students often make the mistake of thinking that to 'show' something requires less rigour than to 'prove' it.

Problem 0.4. Prove parts i and iii of Theorem 0.3.

In Problem 0.2 we asked how many graphs there are on 4 vertices. In this case, we counted 6 different graphs with 1 edges each. But these were all isomorphic. Usually we want to know how many graphs there are 'up to isomorphism', that is, how many isomorphism classes there are. So answer the question properly now!

Problem 0.5. How many non-isomorphic graphs are there on 4 vertices?

Occasionally we still want to count isomorphic graphs as distinct. When we do, we will ask for the number of *labelled* graphs.

As we usually consider graphs to be the same if they are isomorphic, we would like to decide if two graphs are isomorphic. Deciding if two graphs are isomorphic is usually pretty hard (this statement can be made rigourous). But if we are lucky, there are easy ways to prove that they are not.

For example, if two graphs have a different number of vertices, or a different number of edges, they are not isomorphic. Such properties, which are 'preserved by isomorphism', are called graph properties. Graph Theory is the study of these properties.

Some Common Graphs

We generally now consider graphs to be the same if they are isomorphic, and take blatent liberties with the names of their vertices. As such, we define some common graphs, but will apply these names to any graph that is isomorphic.

- The complete graph on n vertices, K_n, has $V(K_n) = \{v_1, v_2, \dots v_n\}$ and $E(K_n) = \{v_i v_j \mid 1 \leq i < j \leq n\}$.

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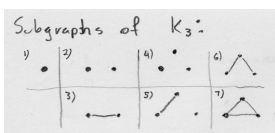
- The n-path, or path of length n, P_n, has $V(P_n) = \{v_0, v_1, \dots v_n\}$ and $E(P_n) = \{v_i v_{i+1} \mid i = 0, \dots, n-1\}$.
- The n-cycle, or cycle of *girth* n, C_n, has $V(C_n) = \{v_1, v_2, \dots v_n\}$ and $E(C_n) = \{v_i v_{i+1} \mid i = 1, \dots, n-1\} \cup \{v_n v_1\}$.
- The complete bipartite graph, $K_{m,n}$, has $V(K_{m,n}) = \{a_1, \dots a_m\} \cup \{b_1, \dots, b_n\}$ and $E(K_{m,n}) = \{a_i b_j \mid i \in [m], j \in [n]\}$.
- The n-cube, Q_n, $V(Q_n)$ is the set of binary strings of length n, vertices u and v are adjacent if they differ in exactly one coordinate.
- The $n \times m$ grid $G_{n \times m}$ is the graph with vertex set $[n] \times [m] = \{(x, y) \mid x \in [n], y \in [m]\}$ in which vertices (x, y) and (x', y') are adjacent if
  i. $x = x'$ and $|y - y'| = 1$, or
  ii. $y = y'$ and $|x - x'| = 1$.

Problem 0.6. Draw the 4×5 grid $G_{4 \times 5}$.

Subgraphs

Given a graph G, another graph G' is a *subgraph* of G, written $G' \leq G$ if $V(G') \subset V(G)$ and $E(G') \subset E(G)$.

A subgraph G' of G is *spanning* if $V(G') = V(G)$, or *induced* if $E(G') = \{uv \in E(G) | u, v \in V(G')\}$. A subgraph G' of G is a *proper* subgraph of G if $E(G')$ is a proper subset of $E(G)$.



Example 0.4. Here are the 7 (upto isomorphism) subgraphs of K_3. All but the 7th are proper, numbers $4, 5, 6$ and 7 are spanning and numbers $1, 3$ and 7 are induced.

Problem 0.7. How many subgraphs are there of K_4? How many are proper? ...induced? ...spanning? ...spanning, induced and proper?

Observe that we have already counted the number of spanning subgraphs of K_4. This is just the number of graphs on 4 vertices.

Problem 0.8. How many subgraphs of K_5 have exactly 5 edges.

Problem 0.9. How many subgraphs of $G_{4 \times 16}$ are isomorphic to C_4? ... to C_5? ... to C_6?

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1 Counting

This section draws from Sections 1.1 - 1.4 of the text. It is understood that the students have seen basic counting techniques, so we only review these topics quickly. Many of what would be examples are posed as problems.

Example 1.1. Take the 9×9 grid $G_{9\times 9}$. Colour a vertex (x, y) red if $x = 3, 6$, or 9, and blue if $y = 3, 6$, or 9. (If a vertex is coloured red and blue then it is now purple, and no longer red or blue.)

We are interested in counting the number of red, blue, purple, and uncoloured vertices. Let R, B, P, and U represent the sets of these vertices respectively.

Clearly we could just count, and find that $|R| = 18$, $|B| = 18$, $|P| = 9$ and $|U| = 38$. There are some basic techniques that allow us to simplify this task.

1.1 Basic Counting Techniques:

The Rule of Sum (Divide and Conquer): If we can break a set A into smaller sets $A_1, \ldots, A_d$ such that every element is in exactly one set, then

$$|A| = |A_1| + \cdots + |A_d|.$$

For example, to count $|R|$ we could break the set $V = V(G_{9\times 9})$ into sets $V_1, \ldots, V_9$ where

$$V_i = \{(x, y) \mid x = i\}.$$

For each i we have $|V_i| = 9$, and we can easily count $R_i = R \cap V_i$ for each i. If $i = 3, 6$, or 9 then $|R_i| = 6$ and otherwise $|R_i| = 0$. So $|R| = 0 + 0 + 6 + 0 + 0 + 6 + 0 + 0 + 6 = 18$.

Example 1.2. We should have used the rule of sum to count the number of spanning subgraphs of K_4. First we count the number with no edges. There is only 1. With one edge, there is 1. With two edges, there are 2; either the edges share a vertex, or they don't. There are 3 with three edges. We now have to count those with four, five or six edges. There are $2, 1$ and 1 respectively. By the rule of sum, there are

$$1 + 1 + 2 + 3 + 2 + 1 + 1 = 13$$

spanning subgraphs of K_4.

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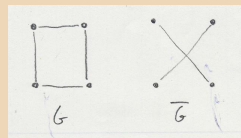
Note

In the above example, how did we count the number of graphs that have four, five, or six edges? We had already counted them.

Given a graph G, the *complement* of G, denoted $\overline{G}$ is the graph with $V(\overline{G}) = V(G)$ and edges defined by

$$E(\overline{G}) = \{uv \mid uv \notin G\}.$$



Problem 1.1. If a graph G has n vertices and m edges, how many edges are there in the complement $\overline{G}$.

Problem 1.2. A graph G is self-complementary if $G \cong \overline{G}$. How many edges are there in a self-complementary graph on n vertices?

Problem 1.3. Show that for graphs G_1 and G_2 with the same number of vertices, $G_1 \cong G_2$ if and only if $\overline{G_1} \cong \overline{G_2}$.

It follows that the number of spanning subgraphs of K_4 with 4 edges is the same as the number with 2 edges.

Symmetry (Counting by bijection)

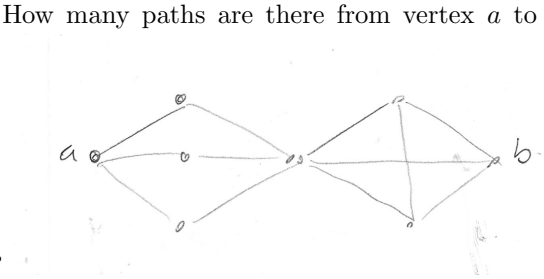
If we have a bijection of V taking a set A to a set A', then $|A| = |A'|$.

For example, the function $f : V \to V : (x, y) \mapsto (y, x)$ is a bijection taking R to B. So $|B| = |R| = 21$.

The Rule of Product: One good way of counting elements of a set is considering the process of choosing an element of the set. If the process can be broken down into independent steps, we can simply count the ways of completing each step.

A vertex is in P if it was painted red and then painted blue. A vertex (x, y) can be painted red in 3 ways: x can be $3, 6$, or 9. It can then be painted blue in 3 ways: y can be $3, 6$, or 9. So there are $3 \cdot 3 = 9$ ways to paint a vertex purple. This shows that $|P| = 9$.

Problem 1.4. How many paths are there from vertex a to vertex b in the following graph?



Counting the Complement: Sometimes it is easy to count a bigger set and then remove items not in the wanted set.

For example, to count the uncoloured vertices U of $G_{9\times 9}$, it might be easier

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count the set V of all vertices and then remove those that are coloured:

$$|U| = |V| - |R| - |B| - |P| = 81 - 18 - 18 - 9 = 38.$$

The *complement* $\overline{A}$ of a subset $A \subset X$ is the set

$$\overline{A} = \{x \in X \mid x \notin A\}$$

of elements not in A. So this is called counting the complement.

Problem 1.5. Let S be the set of two digit integers not divisible by 3:

$$S = \{10, 11, 13, 14, 16, 17, 19, \ldots, 97, 98\}.$$

Find $|S|$.

A graph is *acyclic* if it contains no cycles as subgraphs.

Problem 1.6. How many graphs on 5 vertices with exactly 4 edges, are acyclic?

Note

There are no acyclic graphs on n vertices with n or more edges. Can you show this? This is more of a graph question than a counting question. We will prove it later when we look at trees.

1.2 Permutations

Permutations of a set

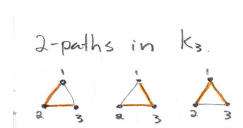
A *permutation* of a set is an arrangement of the items in a line. So $1, 2, 3$ and $3, 1, 2$ are different permutations of the set $\{1, 2, 3\}$. Using the rule of product, one can show that there are $n!$ Indeed, there are n ways to choose the first element, and then $n - 1$ ways to choose the second, etc.

An *r-permutation* or a *size-r permutation* of set is an arrangement of r items of the set in a line. Again by the rule of product, one can show that there are

$$P(n, r) := \frac{n!}{(n-r)!}$$

r-permutations of an n element set.

Problem 1.7. How many different r-paths are there in a labelled K_n?



Problem 1.8. How many subpaths of length n are there in Q_n from the vertex $(0, 0, \ldots 0)$ to the vertex $(1, 1, \ldots, 1)$?

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Permutations of a multiset

A *multiset* is a collection of elements that are not necessarily distinct. The multiset $\{1, 2, 2, 2\}$ is different from $\{1, 2\}$, though as sets, they would be the same. Counting permutations of a multiset adds a slight complication to the task of counting permutations of a set: certain permutations are indistinguishible, so we consider them the same. The multiset $\{1, 2, 2, 2\}$ only has 4 distinguishible permutations:

$$\{(1, 2, 2, 2), (2, 1, 2, 2), (2, 2, 1, 2), (2, 2, 2, 1)\}.$$

This is less than the $4! = 24$ permutations of a 4-element set.

We could get this by counting the elements of the multiset as distinct, in $4!$ ways, and then observing that each distinguishible permutation of the multiset is counted $3!$ times, one for each permutation of the repeated elements. So there are $4 = 4!/3!$ distinguisible permutations of the multiset. More generally we have...

Rule: In a multiset containing n_i *indistinguisible* elements of type i for $i = 1, \ldots, r$, the number of *distinguishible* permutations is

$$\frac{(n_1 + n_2 + \cdots + n_r)!}{n_1! \cdot n_2! \cdot \cdots \cdot n_r!}.$$

When we talk of permutations of a multiset, it is understood that we are talking of distinguishible permutations, so we will generally drop the 'distinguishible.'

Problem 1.9. Find the number of permutations of the letters MISSISSIPPI.

Problem 1.10. Find the number of permutations of the letters ABCDEFGH in which the letters of the word CAGE occur in this order. (So **CA**BH**GE**DF is a valid permutation, but **GE**BH**CA**DF is not.)

Problem 1.11. Find the number of permutations of the letters ABCDEFGH in which the word CAGE doesn't occur (So **CA**BH**GE**DF is valid, but BH**CAGE**DF is not.)

Problem 1.12. How many ways are there to colour the vertices of a path of length 7, (so having 8 vertices,) so that three vertices are red, two are blue, and three are yellow?

Problem 1.13. Count the number of paths in the plane from $(0, 0)$ to $(4, 9)$, using only steps that go one unit up or 1 unit to the left. (Hint See Text Example 1.14.)

Problem 1.14. How many paths in the plane from $(0, 0)$ to $(8, 12)$ touch a vertex on the line $y = x + 3$. (This is hard unless you use a clever trick with the right reflection of the plane.)

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Circular Permutations

A *circular permutation* of a set is an arrangement of the elements in a circle. Two such arrangements are considered the same if you can get from one to the other by rotating the circle. These are the permutations we are considering in the next problem.

Problem 1.15.

(a) How many ways can 6 people be seated around a circular table. (That is, how many circular permutations are there of a 6 element set?)

(b) How many ways can 3 men and 3 women be seated around a circular table so that men and women alternate.

(c) How many ways can 3 couples be seated around a table so that the couples are seated together.

Problem 1.16. How many different r-cycles are there in a K_n? (Hint: In K_3 there is one 3-cycles, in K_4 there are 4.)

Problems from the Text

Section 1.1 and 1.2, page 11-12: 1,3,10,11,13,21,22,24,34,35,37

'The text' refers to Grimaldi 'Discrete and Combinatorial Mathematics (5th edition)'.

1.3 Combinations without replacement

An r-combination of an n-element set is a choice of r elements from the set. If the r-elements must be distinct, there are

$$\binom{n}{r} = P(n,r)/r! = \frac{n!}{(n-r)!r!}$$

of them, (they are permutations in which order is disregarded.) The notation $\binom{n}{r}$ is read 'n choose r', and is called a *binomial coefficient* because of the following theorem.

Theorem 1.3 (Binomial Theorem)**.**

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^{n-i} y^i$$

Proof. Indeed, $(x+y)^n$ is the product of n factors $(x+y)$. Every summand in the expansion of $(x+y)^n$ corresponds to a choice of either x or y from each of these terms. The summand $x^{n-i}y^i$ will occur once for every way we can choose i of the n terms to choose y from. This is $\binom{n}{i}$ ways. □

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Here is the typical problem.

Example 1.4. A pizza place has 16 different toppings. On a deluxe pizza you can choose 6 different toppings. How many different deluxe pizzas are there?

Solution

This is pretty straight forward: $\binom{16}{6}$.

See if you can deal with the following complications of this problem.

Problem 1.17. A pizza place offers the following toppings: 4 kinds of cheese, 7 kinds of vegetables, and 5 kinds of meat. How many different pizzas have

i. 2 cheeses, 4 vegetables and 3 meats (all different).

ii. 6 different toppings at least 3 of which are meat.

iii. 6 different toppings at least 3 of which are cheese.

Some problems can be dealt with as permutations or combination problems.

Example 1.5. How many ways can we split 20 people into 4 teams A, B, C, D of 5 people each. (Teams are distinguishable.)

Solution

We can view this as a permutation problem or a combination problem. We can choose 5 for the first team, 5 for the second, etc., so there are

$$\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}.$$

Alternately, we can arrange 5 of each of A, B, C, D among 20 people:

$$\frac{20!}{5!5!5!5!}$$

You can show that these quantities are the same.

Example 1.6. How many ways can you arrange the letters MISSISSIPPI such that there are no adjacent I's.

Solution

We can do this using combinations and permutations together. First arrange the letters MSSSSPP in $7!/(4!2!)$ ways. Given such an arrangement: SMSPSSP, there are 8 spaces in which we can put one I: -S-M-S-P-S-S-P-. There are $\binom{8}{4}$ ways to place the 4 I's in these spaces, so the total number of arrangements is

$$\frac{7!}{4!2!}\binom{8}{4}.$$

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Problem 1.18. What is the coefficient of x^3y^3 in

i. $(x+y)^6$?

ii. $(2x+y)^6$?

Problem 1.19. What is the coefficient of x^{50} in $(x^6+x^7+x^8+\dots)^6$?

Problems from the Text

Section 1.3: 3,5,6,7,9,12,18,21,23
(For Problem 9, a byte is a string of eight '0's or '1's.)

1.4 Combinations with replacement

Example 1.7. A doughnut store has kruellers, fritters and honey glazed doughnuts. How many ways can you select a dozen?

There is a one-to-one correspondence between the different dozens and the arrangements of 12 stars: '$*$' and 2 dividers: '$|$'. The arrangement

$$**\,|\,*\,*\,*\,*\,*\,*\,*\,|\,*\,*\,*$$

partitions the 12 stars into three parts. This corresponds to the dozen made up of 2 kruellers, 7 fritters, and 3 honey glased doughnuts. Thus there are $\binom{12+2}{2}$ differnt dozens.

It is clear that in our example, we also counted the solutions of

$$x_1+x_2+x_3=12$$

where x_1, x_2, x_3 are non-negative integers. The value of x_i in a solution corresponds to the number of '$*$'s in the ith part.

Here are a couple of variations on the question.

Hint

(Problem 1.20) One can do this by counting the non-negative integer solutions of $x_1+x_2'+x_3=10$ (where $x_2'=x_2-2$).

Problem 1.20. How many solutions are there to the equation $x_1+x_2+x_3=12$ where all x_i are non-negative integers and $x_2 \geq 2$?

Hint

(Prob. 1.21) One can do this by counting the non-negative integer solutions of $x_1+x_2+x_3+x_4=12$, (where x_4 is called a slack variable.)

Problem 1.21. How many solutions are there to the equation $x_1+x_2+x_3 \leq 12$ where all x_i are non-negative integers.

Problem 1.22. A composition of an integer n is an ordered sum of positive integers totalling to n. (So 8, $7+1$, $1+7$ and $1+1+1+1+1+1+1+1$ are all different compositions of 8.) How many compositions are there of 8?

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Problem 1.23. How many ways can seven 'H's and nine 'T's be arranged in a line so that there exists exactly four strings of consecutive 'T's?

Problem 1.24. How many ways are there to colour the vertices of K_{12} with three colours? What if we consider two colourings the same when we can get from one to the other by an isomorphism of K_{12} to K_{12}? How about for $\overline{K_{12}}$?

The above problem becomes quite difficult for C_{12}. See Section 16.13 of the text if you are interested.

Problems from the Text

Sect 1.4: 1,3,7,8,10,13,18

These are dice:

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2 Symbolic Logic

This draws from Chapter 2 of the text.

Recall that an integer is prime if it is greater than one, and its only positive factors are one and itself. The following is a (sketch of a) well known proof of the fact that there are infinitely many primes.

Proof. Assume that there are only finitely many primes. Then we may list them: $p_1, p_2, \ldots, p_n$. Let $P = (p_1 \cdot p_2 \cdot \cdots \cdot p_n) + 1$. P is greater than all of the p_i so it is not prime. Since it is not prime, it is divisible by some prime, so can be written $P = Qp_i$ for some p_i. But then p_i divides P and $P - 1$ so divides $P - (P - 1) = 1$. This is impossible, because $p_i \geq 2$. Our assumption that there are only finitely many prime must, therefore, be false. So there are infinitely many primes. $\square$

We wanted to prove a statement 'S is true.' but we instead assumed 'S is false' and showed this was inconsistent. This is called a 'proof by contradiction', and is a common proof technique. Similarily to prove an implication 'if a then b', a useful approach is to show that 'if b is not true, then a is not true'. Why is this okay?

Symbolic logic is useful in deciding if two approaches to a proof are equivalent. We look at symbolic logic with a view to proof, techniques, but take it further than this.

2.1 Basic Connectives and Truth Tables

Our basic structure in this section is a *statement* s, and looks something like the following example.

$$s : (p \wedge q) \rightarrow \neg p \vee r$$

In this example, the letters p, q and r are called *primitive statements* or primitives, and the symbols $\wedge$, $\vee$, $\rightarrow$ and $\neg$ are called *(logical) connectives.* Statements that are not primitive, are called *compound statments.*

Primitive statements stand for simple mathematical statements, such as '8 is prime', or '2 divides 7', which can be either true or false. Depending on the truth value of the primitives a statement involving them may be true or false. If p is the statement '8 is prime.' then $\neg p$, read 'not p', is the statement '8 is not prime.' In this case p is false and $\neg p$ is true.

But we are not so interested in the truth of the primitives p. We are more interesed in statments such as

$$s : p \vee \neg p$$

read 'p or not p,' that are true whether or not p is true.

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So we can kind of view primitives p and q as variables. But we are careful to use the word 'primitive', because latter we will see something called an *open statement,* written $s(x)$, in which there really is a variable x.

Basic Connectives

Given primitives statments p and q we can construct compound statements using any of the following logical connectives. 'And' denoted $\wedge$; 'or' denoted $\vee$; 'xor' or 'exclusive or' denoted $\underline{\vee}$; 'implies' denoted $\rightarrow$; and 'biconditional' or 'if and only if' or 'iff' denoted $\leftrightarrow$.

Statements can be built up from other statements. For example, where p is $1+1=3$ and q is $1+1=2$, $p \vee q$ is the statement that '$1+1$ is 3 or 2', and is true because at least one of p and q was true.

The truth value, that is, the truth or falsity, of a compound statement depends only on the truth value of the primitives in it. The above connectives are thus rigourously defined by the following table, where we denote truth by 1 and falsity by 0.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \underline{\vee} q$	$p \rightarrow q$	$p \leftrightarrow q$
0	0	1	0	0	0	1	1
0	1	1	0	1	1	1	0
1	0	0	0	1	1	0	0
1	1	0	1	1	0	1	1

This table is called a *truth table.* Truth tables can be used to determine the value of any compound statement from the value of their component primitives. For more complicated statements, we build up the truth table in several steps. The general approach is to approach a statement beginning with 'not' and then bracketed statements, and then working left to right.

Problem 2.1. Make a truth table for the statement $(\neg p \vee q) \rightarrow r$.

Solution

p	q	r	$\neg p$	$\neg p \vee q$	$(\neg p \vee q) \rightarrow r$
0	0	0	1	1	0
0	0	1	1	1	1
0	1	0	1	1	0
0	1	1	1	1	1
1	0	0	0	0	1
1	0	1	0	0	1
1	1	0	0	1	0
1	1	1	0	1	1

Observe that in the above example, the statement was true in five out of eight of the possible truth assignments of the primitives. One thing we are

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interested in is when a statement is always true, or always false, whatever the truth values of the primitives. From now on, we treat primitives as variable, which can be true or false. Though they are true or false, we don't know what they are.

Definition 2.1. A *tautalogy* is a statement that is always true. A *contradiction* is a statement that is always false.

Example 2.2. The statement $p \vee \neg p$ is a tautology and the statement $p \wedge \neg p$ is a contradiction.

Problem 2.2. Using a truth table, decide whether or not the statement $(p \to (q \to p))$ is a tautology or a contradicion. Can you argue this without a truth table?

One sees that if a statement has n primitives, then the truth table will have 2^n rows. Three primitives is about the most we want to deal with. From here the theory of logical systems that we start to develop will allow us to decide if a statement is a truth or contradiction through more efficient means.

Remark 2.3. Denoting truth by 1 and falsity by 0 exemplifies the analogy between the logic algebra that we are developing, and a boolean algebra. We could further view $\underline{\vee}$ and $\wedge$ as analogous to addition and multiplication. We won't follow this analogue any more, but if you are familiar with boolean algebras, it might be a useful analogue.

Problems from the Text

Sect 2.1: 1,3,5,6,8,9,10,11,13

2.2 Logical Equivalence

Problem 2.3. Construct truth tables for the statements '$p \to q$' and '$\neg p \vee q$'.

For every truth assignment of the primitives p and q, these two compound statements have the same truth value.

Definition 2.4. Statements s_1 and s_2 are *logically equivalent* if s_1 is true if and only if s_2 is true. We denote this $s_1 \iff s_2$. If p is a tautalogy, we write $p \iff T_0$ and if it is a contradiction, we write $p \iff F_0$.

Remark 2.5. The notations '$s_1 \leftrightarrow s_2$' and '$s_1 \iff s_2$' mean different things. The first is a logical statement- it may be true or false, and writing it down we are making no assertion about its truth value. The second, '$s_1 \iff s_2$', says that the first, '$s_1 \leftrightarrow s_2$', is a tautalogy.

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It is easy to show (though not always easy to determine) that non-equivalent statements are non-equivalent. One only has to provide one line of the truth table- a true assignment of the primitives in which the statements differ.

Example 2.6. Show that $p \wedge (\neg p \to q)$ is not a tautalogy.

Solution

Where $p = 0$ and $q = 0$ we have $\neg p = 1$ so $(\neg p \to q) = 0$ so $p \wedge (\neg p \to q) = 0 \wedge 0 = 0$. Thus the statement is not a tautalogy.

To prove two statements are equivalent, we can always use a truth table, but we would like to avoid doing this too often. So we build up a list of equivalences that we can use over and over.

Above, we saw that

$$p \to q \iff \neg p \vee q$$

so '$\to$' can be expressed in terms of '$\neg$' and '$\vee$'. In the same way, we have

$$\begin{aligned} p \leftrightarrow q \quad &\iff \quad (p \to q) \wedge (q \to p) \\ &\iff \quad (\neg p \vee q) \wedge (\neg q \vee p) \\ &\iff \quad (\neg p \wedge \neg q) \vee (p \wedge q) \end{aligned}$$

and

$$p \underline{\vee} q \iff (p \vee q) \wedge \neg(p \wedge q).$$

Often we will use these equivalences to reduce compound statements to statements using only $\neg$, $\vee$ and $\wedge$.

Problem 2.4. Convince yourself of the logical equivalences given above. Prove that $\wedge$ can also be expressed in terms of '$\neg$' and '$\vee$'.

There are many other useful logical equivalences

Problem 2.5. Use a truth table to show that

$$\neg(p \wedge q) \iff (\neg p \vee \neg q).$$

This is called DeMorgan's law.

We will frequently use the following Laws of Logic.

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1	$\neg\neg p \iff p$	Double Negation
2	$\neg(p \vee q) \iff \neg p \wedge \neg q$ $\neg(p \wedge q) \iff \neg p \vee \neg q$	Demorgan
3	$p \vee q \iff q \vee p$ $p \wedge q \iff q \wedge p$	Commutativity
4	$p \vee (q \vee r) \iff (p \vee q) \vee r$ $p \wedge (q \wedge r) \iff (p \wedge q) \wedge r$	Associativity
5	$p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$	Distributivity
6	$p \vee p \iff p$ $p \wedge p \iff p$	Idempotence Idempotence
7	$p \vee F_0 \iff p$ $p \wedge T_0 \iff p$	Identity
8	$p \vee \neg p \iff T_0$ $p \wedge \neg p \iff F_0$	Inverse
9	$p \vee T_0 \iff T_0$ $p \wedge F_0 \iff F_0$	Domination
10	$p \vee (p \wedge q) \iff p$ $p \wedge (p \vee q) \iff p$	Absorption
(11)	$p \to q \iff \neg p \vee q$	Implication

Problem 2.6. Read through each of the laws of logic, and convince yourself that they are true. Prove your favourite one using a truth table. Prove that it is your favourite one. Prove that it doesn't pay to choose favourites.

Substitution Rules

On top of the laws of logic, we have some substitution rules which allow us to use them more easily.

- If a statement s is a tautalogy, and every occurence of some primitive p in s is replaced by another statement t, then the resulting statement is a tautology.
- If $s \iff t$, you can always replace s with t in statement without changing the truth values.

Example 2.7. The statement $s : (p \to q) \leftrightarrow (\neg p \vee q)$ is a tautology. Replacing p with $(\neg r \wedge t)$ yields that

$$((\neg r \wedge t) \to q) \leftrightarrow (\neg(\neg r \wedge t) \vee q)$$

is also a tautology.

Example 2.8. By the law of double negation we have:

$$(p \to q) \to \neg p \iff (\neg\neg p \to q) \to \neg p.$$

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Here is a more meaty example of the use of the Laws of Logic and the Rules of substitution.

Example 2.9. Show that $(\neg p \to q) \to r \iff (p \to r) \wedge (q \to r)$.

Solution

	$(\neg p \to q) \to r$	
$\iff$	$(\neg\neg p \vee q) \to r$	Implication
$\iff$	$(p \vee q) \to r$	Double Negation
$\iff$	$\neg(p \vee q) \vee r$	Implication
$\iff$	$(\neg p \wedge \neg q) \vee r$	DeMorgan
$\iff$	$r \vee (\neg p \wedge \neg q)$	Commutativity
$\iff$	$(r \vee \neg p) \wedge (r \vee \neg q)$	Distributivity
$\iff$	$(p \to r) \wedge (q \to r)$	Implication

There are a couple more statements that are related to the implication $p \to q$ which get special names as they are used often as proof techniques.

The *contrapositive* of $p \to q$ is $\neg q \to \neg p$. The *converse* of $p \to q$ is $q \to p$. The *inverse* of $p \to q$ is $\neg p \to \neg q$.

Problem 2.7. Show that an implication is logically equivalent to its contrapositive, but not to its converse or inverse. What does this tell you of the relation between the converse and the inverse of an implication?

In proving an implication, one common technique is to prove the contrapositive. When we prove an implication, we often comment on whether or not the converse is true.

Problem 2.8. Where p is the sentence 'It is raining', and q is the sentence 'I skip class', what are the English interpretations of $p \to q$, its contrapositive, its inverse, and its converse.

Let's finish this section with another example of proving the equivalence of statements using our laws of logic.

Example 2.10. Negate and simplify the statement $p \to (\neg q \wedge r)$.

Solution

Leaving the word 'simplify' a little vague, we proceed:

	$\neg(p \to (\neg q \wedge r)$	
$\iff$	$\neg(\neg p \vee (\neg q \wedge r)$	Using $s \to t \iff \neg s \vee t$
$\iff$	$\neg\neg p \wedge \neg(\neg q \wedge r)$	DeMorgan
$\iff$	$p \wedge (\neg\neg q \vee \neg r)$	Double Negation then DeMorgan
$\iff$	$p \wedge (r \to q)$	Using $s \to t \iff \neg s \vee t$

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Observe that we used Double Negation and DeMorgan together in one step, and then in the next step, we used Double Negation without even mentioning it. We allow such shortcuts if it doesn't hinder reading. But don't take them too far.

The text finished this section with some examples of how a logical statement can be modelled by an electronic circuit. We won't look at it, but it you are so inclined, it is an interesting way to visualise the equivalence of statements.

Problems from the Text

Sect 2.2: 1,3,5,7,9,11,13,15,17,19

2.3 Inference Rules

Logical Implication

Definition 2.11. An *argument* consists of a conjunction of statements :

$$p_1 \wedge p_2 \wedge \cdots \wedge p_d$$

called the *premises* and a *conclusion* q.

We write an argument as

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_d] \to q.$$

It is called true or *valid* if it is a tautalogy, that is, if when all the premises are true, then the conclusion is true.

Example 2.12. Is the following argument valid?

$$[(p \to r) \wedge (\neg q \to p) \wedge \neg r] \to q$$

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Solution

Again, we can use a truth table.

p	q	r	$s_1 : p \to r$	$s_2 : \neg q \to r$	$s_3 : \neg r$	$[s_1 \wedge s_2 \wedge s_3] \to q$
0	0	0	1	0	1	
0	0	1	1	1	0	
0	1	0	1	1	1	1
0	1	1	1	1	0	
1	0	0	0	0	1	
1	0	1	1	1	0	
1	1	0	0	1	1	
1	1	1	1	1	0	

There is only one row in which all of the premises are true, and in this case the conclusion is also true, so the argument is valid.

Example 2.13. Is the argument $[(p \to q) \wedge q] \to p$ valid?

Solution

No! (Don't you dare.) If $q = 1$ and $p = 0$ then $(p \to q) \wedge q$ is 1, but $p = 0$.

Example 2.14. Is the argument $[p \vee \neg p] \to q$ valid?

Indeed it is, as the premises are a contradiction. In this case we say that the argument is *vacuoulsy true*. 'Vacuous' means 'empty', like a vacuum. It is saying, 'sure the argument is true, but this is not really saying anything'.

Definition 2.15. If p and q are statements such that $p \to q$ is a tautology, then we say p *logically implies* q and write $p \Rightarrow q$

Note that $p \to q$ can be true or false, but $p \Rightarrow q$ means $p \to q$ is a tautology. So '$p \iff q$' is the same as '$p \Rightarrow q$ and $q \Rightarrow p$'.

Problem 2.9. Use a truth table to prove that $[(p \to q) \wedge p] \Rightarrow q$.

The logical implication in the above problem is a *inference rule* called *Modus Ponens*. Having proved it, we use it in proving the validity of other arguments.

Example 2.16. To prove the implication $[(p \to q) \wedge \neg q] \Rightarrow \neg p$ we start with the premises, and work to the conclusion.

$$\begin{array}{rcl} & (p \to q) \wedge \neg q & \\ \iff & (\neg q \to \neg p) \wedge \neg q & (contraposative) \\ \Rightarrow & \neg p & (ModusPonens). \end{array}$$

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It will make an argument clearer if we write it in a table as follows.

1	$p \to q$	premise (given)
2	$\neg q$	premise
3	$\neg q \to \neg p$	contraposative of 1
$\therefore$	$\neg p$	2 and 3 and Modus Ponens

The symbol $\therefore$ is read 'therefore', and denotes that we have arrived at a conclusion. The implication above is another inference rule called *Modus Tollens*. We have several more inference rules:

	Premises	Conclusion	Name
1	$p \wedge (p \to q)$	q	Modus Ponens
2	$(p \to q) \wedge (q \to r)$	$p \to r$	Syllogism
3	$(p \to q) \wedge \neg q$	$\neg p$	Modus Tollens
4	$(p) \wedge (q)$	$p \wedge q$	Conjunction
5	$(p \vee q) \wedge \neg p$	q	Disjunctive Syllogism
6	$(\neg p \to F_0)$	p	Contradiction
7	$p \wedge q$	p	Conjunctive Simplification
8	p	$p \vee q$	Disjunctive Amplification
9	$(p \wedge q) \wedge [p \to (q \to r)]$	r	Conditional Proof
10	$(p \to r) \wedge (q \to r)$	$(p \vee q) \to r$	Proof by cases
11	$(p \to q) \wedge (r \to s) \wedge (p \vee r)$	$q \vee s$	Constructive Dilemma
12	$(p \to q) \wedge (r \to s) \wedge (\neg q \vee \neg s)$	$(\neg p \vee \neg r)$	Destructive Dilemma

These will help us prove logical implications.

Example 2.17. Show the argument $[(p \to q) \wedge \neg q \wedge \neg r] \to \neg(p \vee r)$ is valid.

Solution

1	$p \to q$	premise
2	$\neg q$	premise
4	$\neg p$	1 and 2 and Modus Tollens
5	$\neg r$	premise
6	$\neg p \wedge \neg r$	conjunction of 4 and 5
$\therefore$	$\neg(p \vee r)$	6 DeMorgan

Problem 2.10. Prove $[(p \to q) \wedge (r \to \neg q) \wedge r] \Rightarrow \neg p$.

Problem 2.11. Prove $[(p \to (q \to r)) \wedge (p \vee s) \wedge (t \to q) \wedge \neg s] \Rightarrow (\neg r \to \neg t)$.

To show that an argument is false, we need only to give a counterexample.

Example 2.18. To see that

1	$p \to q$
2	$\neg p$
$\therefore$	$\neg q$

is an invalid argument, let $p = 0$ and $q = 1$. Then premises 1 and 2 are true, but the conclusion is false.

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Common Proof Techniques

Two common proof techniques that we get from our inference rules are 'Proof by Contradiction' and 'Indirect proof.' Proof by Contradiction is a proof that uses the rule of contradiction. For this we introduce a false premise, the negation of our conclusion, and show it leads to falsity.

Example 2.19. We show that

1	$(s \wedge h) \to r$
2	$r \to c$
3	$\neg c$
$\therefore$	$\neg s \vee \neg h$

is valid. Indeed, taking 1 - 3 as premises, we continues as follows

4	$\neg(\neg s \vee \neg h)$	contradictory premise
5	$s \wedge h$	4 and DeMorgan
6	r	1 and 5 and MP (Modus Ponens)
7	c	2 and 6 and MP
8	$c \wedge \neg c$	3 conjunction 7
9	F_0	8 and Inverse Law
$\therefore$	$\neg s \wedge \neg h$	4 and 9 and contradiction

Since $(p \to (q \to r)) \iff ((p \wedge q) \to r)$, to show

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_n] \to (q \to r)$$

it is enough to show

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge q] \to (r).$$

This is called an indirect proof.

Example 2.20. We prove the Rule of Syllogism (see the table) by indirect proof.

1	$p \to q$	premise
2	$q \to r$	premise
3	p	indirect premise
4	q	1,3, MP
$\therefore$	r	2,4 MP

Thus by indirect proof, we get the rule of Syllogism.

Problem 2.12. Prove

1	$p \to (q \to r)$
2	$\neg q \to \neg p$
$\therefore$	$p \to r$

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Problems from the Text

Sect 2.3: 1,3,9,11

2.4 Quantifiers

Recall that statements such as

$$p(x) : x \text{ is a multiple of } 3,$$

and

$$q(x, y) : x > y,$$

are not **statements**. Because of the variable, their truth values cannot be determined. However, when their variables are replaced with numbers, the become statements.

Definition 2.21. An *open statement* is a sentence that

- contains one or more variable,
- becomes a statement when the variables are replaced with certain allowable choices.

Example 2.22. Where $p(x)$ is the open statement above, $p(6)$ is the true statement '6 is a mutliple of 3', while $p(5)$ is the false statement '5 is a mutliple of 3'.

There is another way to change a open statement into a statement.

Definition 2.23. The allowable choices for the variables of an open statement are called the *universe* or *universe of discourse*, of the open statement. The symbol $\forall$, read as 'for every', is the *universal quantifier*. The symbol $\exists$, read as 'for some', is the *existential quantifiers*. They *quantify* an open statement $r(x)$ with universe U, as follows.

- $\forall x[r(x)] \iff \bigwedge_{x \in U} r(x)$
- $\exists x[r(x)] \iff \bigvee_{x \in U} r(x)$

That is, '$\forall x[r(x)]$' is the statement that '$r(x)$ is true for **all** $x \in U$,' and '$\exists x[r(x)]$' is the statement that '$r(x)$ is true for **some** $x \in U$.' (We ignore the problem that these might be a sentences consisting of infinitely many primitives.)

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Note

Often, the universe of an open statement does not matter, or can be assumed from 'what is reasonable'. Certainly to interpret $p(6)$ as the statement '6 is a multiple of 3', we don't need to know the universe. But for quantified statements, it is usually important. Indeed, where $p(x)$ is as above $\forall x[p(x)]$ is false if the universe is $U = \mathbb{Z}$, but true if it is $U = 3\mathbb{Z}$.

That said, choosing the universe as $U = 3\mathbb{Z}$ seems a little like trickery. Sometimes even for open statements, if the universe is omitted, we assume it to be the largest reasonable universe. The statement 'x is a multiple of 3' appears very much to be about the integers, it would be reasonable to assume this is the universe.

Example 2.24. Let $q(x, y)$ be the statement $x > y$, with universe $\mathbb{R}$. Then

(a) $\forall x \forall y[q(x, y)]$ is false.

(b) $\forall x \exists y[q(x, y)]$ is true.

(c) $\exists y \forall x[q(x, y)]$ is false.

Hey look! The order of the quantifiers matters.

Problem 2.13. For each of the statements in the previous example, change the universe so that the truth values of the statement changes.

Problem 2.14. Where the universe is $\mathbb{Z}$, let

- $p(x) : x^2 - 5x + 6 = 0$
- $q(x) : x > 0$
- $r(x) : x^2 - 7 \geq 0$.

Determine the truth value of the following statements.

(a) $\exists x[p(x) \wedge \neg q(x)]$

(b) $\exists x[\neg q(x) \rightarrow p(x)]$

(c) $\forall x[p(x) \rightarrow q(x)]$

(d) $\forall x[p(x) \vee (q(x) \wedge r(x))]$

Hint

You can use arithmetic symbols $<$, $\leq$, and $=$.

Problem 2.15. Where the universe of x is the set of students in the class, and the universe of y is a score, let $s(x, y)$ be the open statement:

x got y on the test.

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Express the following statements symbolically.

(a) Some student got 7.5 on the test.

(b) Every student got at least 5 on the test.

(c) Some student got more than 9 on the test.

(d) No student got 8.5 on the test.

(e) Only one student got 0 on the test.

Quantifications of Logical Equivalences

Recall the logical equivalence

$$p \to q \iff \neg p \vee q.$$

It follows by substitution that if $p(x)$ and $q(x)$ are open statements with the same universe, then

$$p(x) \to q(x) \iff \neg p(x) \vee q(x)$$

for any choice of x in the universe.

Thus the following quantified versions of the logical equivalence are also true:

- $\forall x[p(x) \to q(x)] \iff \forall x[\neg p(x) \vee q(x)]$
- $\exists x[p(x) \to q(x)] \iff \exists x[\neg p(x) \vee q(x)]$

Every logical equivalence has similarily quantified versions.

Interaction of Quantifiers and Connectives

One can show the following.

Proposition 2.25. *(a)* $\forall x[p(x) \wedge q(x)] \iff \forall x[p(x)] \wedge \forall x[q(x)]$

(b) $\exists x[p(x) \wedge q(x)] \Rightarrow \exists x[p(x)] \wedge \exists x[q(x)]$

(c) $\forall x[p(x) \vee q(x)] \Leftarrow \forall x[p(x)] \vee \forall x[q(x)]$

(d) $\exists x[p(x) \vee q(x)] \iff \exists x[p(x)] \vee \exists x[q(x)]$

(e) $\neg \forall x[p(x)] \iff \exists x[\neg p(x)]$

(f) $\neg \exists x[p(x)] \iff \forall x[\neg p(x)]$

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Proof. We prove only part (a), and assume a finite universe $U = x_1, \ldots, x_n$ of the variable x.

$$\begin{aligned}
\forall x[p(x) \wedge q(x)] \quad &\iff \quad p(x_1) \wedge q(x_1) \wedge p(x_1) \wedge q(x_1) \wedge \cdots \wedge p(x_n) \wedge q(x_n) \\
&\iff \quad (p(x_1) \wedge p(x_2) \wedge \cdots \wedge p(x_n)) \wedge (q(x_1) \wedge q(x_2) \wedge \cdots \wedge q(x_n)) \\
&\iff \quad (\forall x[p(x)]) \wedge (\forall x[q(x)])
\end{aligned}$$

□

Problem 2.16. Convince yourself that the logical implications from Proposition 2.25 are true. Show that $\Leftarrow$ is not true in (b) and $\Rightarrow$ is not true in (c).

The above rules allow us to prove other logical implications involving the quantifiers.

Example 2.26. Show that $\exists x[p(x) \to q(x)] \wedge \forall x[p(x)] \Rightarrow \exists x[q(x)]$.

Solution

1	$\exists x[p(x) \to q(x)]$	premise
2	$\exists x[\neg p(x) \vee q(x)]$	1 and implication
3	$\exists x[\neg p(x)] \vee \exists x[q(x)]$	2 and Prop 2.25
4	$\neg \forall x[p(x)] \vee \exists x[q(x)]$	3 and Prop 2.25
5	$\forall x[p(x)]$	premise
$\therefore$	$\exists x[q(x)]$	4,5, Modus Tollens.

Problems from the Text

Sect 2.4: 1,7,9,11,18,21

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3 Set Theory

We cover Section 3.1-3.4 of the text; however, we might have rearranged material a bit.

3.1 Sets and Subsets

This mostly agrees with Section 3.1 of the text.

In Chapter 0 we defined *subsets* $S \subseteq T$. We can write our definition in set-builder notation as follows.

$$S \subseteq T \iff \forall x[x \in S \Rightarrow x \in T]$$

The universe here is the set of all elements of any set.

How would you define the statement $A = B$, that the sets A and B are the same set? One way would be

$$\forall x[x \in A \iff x \in B]$$

.

But look at the following logical equivalence:

$$\begin{aligned} \forall x[x \in A \iff x \in B] \quad &\iff \quad \forall x[x \in A \to x \in B] \wedge \forall x[x \in B \to x \in A] \\ &\iff \quad A \subseteq B \wedge B \subseteq A \end{aligned}$$

This rigourously proves the obvious statement that

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

As simple as this seems, we will use it a lot in proofs. It can be considered an alternate definition of $\subseteq$.

A set A is a *proper subset* of B, written $A \subset B$ if $A \subseteq B$ and $A \neq B$.

Problem 3.1. Give an alternate definition of $A \subset B$:

$A \subseteq B$ and $\exists$ _____.

As the basic set definitions have 'quantified open statement' versions, proofs about set theory will be very similar to proofs of logical equivalences and logical implications. Let's see one.

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Theorem 3.1. *Let $A, B, C \subseteq U$.*

i. If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

ii. If $A \subset B$ and $B \subseteq C$ then $A \subset C$.

iii. If $A \subseteq B$ and $B \subset C$ then $A \subset C$.

iv. If $A \subset B$ and $B \subset C$ then $A \subset C$.

Proof. We prove only *i* and *iii*. To prove *i*, assume the premises, that $A \subseteq B$ and $B \subseteq C$. By definition this means that

$$\begin{aligned} & \forall x[x \in A \to x \in B] \wedge \forall x[x \in B \to x \in C] \\ \iff & \forall x[(x \in A \to x \in B) \wedge (x \in B \to x \in C)] \\ \Rightarrow & \forall x[x \in A \to x \in C] \text{ by syllogism.} \end{aligned}$$

This last statement is the definition of $A \subseteq B$.

Now we prove *iii*. Assume that $A \subseteq B$ and $B \subset C$. As $B \subset C$ implies $B \subseteq C$ we have the premises of part *iii*., so we may conclude that $A \subseteq C$. Thus we only have to show that there is some x that is in C but not A. By the premise $B \subset C$ there is some x in C that is not in B. As A is a subset of B, this x is not in A either. So we are done. □

Notice that our proof style differed from part *i* to part *iii*. In *i* we were very notational while in *iii* we relied a lot more on english sentences. Which one is easier for you to read? Mathematics tends to be written as in part *iii*, as it is easier to read. But it is still precise, many words or phrases have more restricted, or different usage in Math than they do in regular conversation. It is good to learn some of these usages.

Note on English

The word 'assume' is usually comes at the beginning of a proof and will be followed, often much later, by

- a conclusion, if we are proving an argument to be valid; or
- a contradiction, when we are doing proof by contradiction.

When the proofs involved between the assumptions and the conclusion/contradiction are very short, we often use 'if' instead of 'assume'. Ex. 'If x is in A then it is also in B, which is false, so x is not in A.'

Problem 3.2. Finish the proof of Theorem 3.1. Try to write it more like the proof of part *iii* above.

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Okay, enough of the boring stuff.

The *powerset* of a set S, written $\mathscr{P}(S)$ or 2^S is the set of all subsets of S. The *cardinality* $|S|$ of a finite set S is the number of elements in the set. For infinite sets such as $\mathbb{Z}$ or $\mathbb{R}$, we say the cardinality is ∞. In other classes you will see that there are various infinite cardinalities, but that won't interest us in this class.

The power set of $\mathscr{P}(S)$ of a set S is the set of all subsets of S.

Problem 3.3. If $|S| = n$, what is $|\mathscr{P}(S)|$?

Solution

One easy way to count $|\mathscr{P}(S)|$ is using the product rule. In choosing an element of $\mathscr{P}(S)$, that is, a subset T of S, we have to choose for each of the n elements x of S whether or not x is in T. There are 2^n ways to do this, so $\mathscr{P}(S) = 2^n$.

Another way to count is to use the sum rule. Counting the 0 element sets, then the 1 element sets, et cetera, we get $|\mathscr{P}(S)| = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}$.

Double Counting

Above we counted $|\mathscr{P}(S)|$ in two ways, getting the identity

$$2^n = \sum_{i=0}^{n} \binom{n}{i}.$$

This technique for finding the identity is an example of what is called *double counting*. This is a common technique for proving combinatorial indentities.

Problem 3.4. Show that this identity also follows from the Binomial theorem.

Problem 3.5. Show that

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$$

by double counting the number of r-combinations of an $n+1$ element set.

Problems from the Text

Sect 3.1: 9,18,21

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3.2 Set Operations and Laws of Set Theory

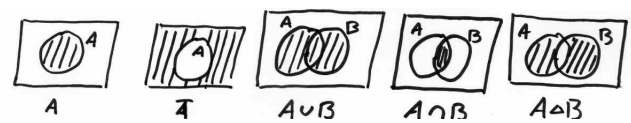
This is Mostly Section 3.2 and 3.3 of the text.

Each of the basic logical connectives defines an operation on sets.

Let A and B be sets with the same universe U, then we have

- the *complement* $\overline{A} = \{x \mid \neg(x \in A)\}$
- the *union* $A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$
- the *intersection* $A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$, and
- the *symmetric difference* $A \triangle B = \{x \mid (x \in A) \veebar (x \in B)\}$.

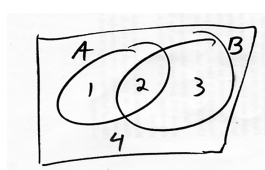
These set operations can be visualised by a *Venn (or circle) Diagram.*



For every law of logic, there is a corresponding law of set theory. One such law is DeMorgan's Law of Set Theory:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

We can prove that this is true by using a Venn diagram as follows.



Example 3.2. Consider an element x in region i of the diagram. Then $x \in A \cup B \iff i \in \{1, 2, 3\}$. So $x \in \overline{A \cup B} \iff x \in \{4\}$. On the other hand $x \in \overline{A} \iff i \in \{3, 4\}$ and $x \in \overline{B} \iff i \in \{1, 4\}$, so $x \in \overline{A} \cap \overline{B} \iff i \in \{4\}$. Thus $x \in A \cup B \iff x \in \overline{A} \cap \overline{B}$.

This is essentially a truth table. (In fact, you can set it up in a table.) From definitions we could prove it as follows, using the DeMorgan law of logic.

$$\begin{array}{rcll}
\overline{A \cup B} & = & \overline{\{x \mid x \in A \vee x \in B\}} & \text{def of union} \\
 & = & \{x \mid \neg(x \in A \vee x \in B)\} & \text{def of complement} \\
 & = & \{x \mid x \notin A \wedge x \notin B\} & \text{DeMorgan} \\
 & = & \{x \mid x \in \overline{A} \wedge x \in \overline{B}\} & \text{def of complement} \\
 & = & \overline{A} \cap \overline{B} &
\end{array}$$

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This seems the better way to do it.

Another set operation is the *relative complement of A in B* defined as

$$A - B = \{x \mid x \in A \vee x \notin B\}.$$

Problem 3.6. Prove that $A - B = A \cap \overline{B}$

Problem 3.7. What is the set $\{x \mid x \in A \to x \in B\}$? Perhaps use the implication law. Draw a Venn diagram to describe it. How does this relate to $A - B$.

As the DeMorgan Law of Set Theory followed immediately from the DeMorgan Law of Logic, we get the Set Laws, set theoretic versions of all of the laws of logic. See the table on page 139 of the text. It is useful to also include the identity from Problem 3.6, which we view as an analogue of the implication law, in this list.

These allow us even more direct proofs of set equivalences which do not require us to apply explicitly to the theory of logic.

Example 3.3. Show that $A - (B \cup C) = (A - B) \cap (A - C)$.

Solution

$$\begin{aligned} A - (B \cup C) &= A \cap \overline{B \cup C} && \text{Implication Law} \\ &= A \cap \overline{B} \cap \overline{C} && \text{DeMorgan} \\ &= A \cap A \cap \overline{B} \cap \overline{C} && \text{Idempotence} \\ &= (A \cap \overline{B}) \cap (A \cap \overline{C}) && \text{Commutativity} \\ &= (A - B) \cap (A - C) && \text{Implication} \end{aligned}$$

Problems from the Text

Sect 3.2: 6,7,8,13,17

3.4 A First Word on Probability

Probability is a convenient way of counting the relative sizes of sets.

Definition 3.4. An *experiment* is an activity with some set of possible outcomes. The *sample space* is the set S of possible outcomes. An *event* is a subset A of the sample space.

Example 3.5. In an experiment we toss two coins. Each coin can come up 'Heads' (H) or 'Tails' (T), so our sample space is $S = \{HH, HT, TH, TT\}$. Some possibles events are

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- $A_1 = \emptyset$.
- $A_2 = \{HT, TH\}$.
- $A_3 = S$.

If an experiment has a sample space in which each outcome is equally likely, the probability $Pr(A)$ of an event A is

$$Pr(A) = |A|/|S|.$$

We write $Pr(a)$ for $Pr(\{a\})$.

Example 3.6. Continuing the previous example, we have that $Pr(A_1) = 0$, $Pr(A_2) = 1/2$, and $Pr(A_3) = 1$.

Example 3.7. Two distinguishable dice are rolled. Find the probability that the rolled numbers sum to a) 7, b) 9, c) 12. d) Find the probability that at least one 2 is rolled.

Solution

The sample space S has 36 elements.
a) $Pr(\{16, 25, 34, 43, 52, 61\}) = \frac{6}{36} = 1/6$
b) $Pr(\{36, 45, 54, 63\}) = \frac{4}{36} = 1/9$
c) $Pr(\{66\}) = 1/36$
d) Let A_1 be the event that the first die is a 2 and let A_2 be the event that the second die is a 2. The we are looking for

$$\begin{aligned} Pr(A_1 \cup A_2) &= \frac{1}{36}|A_1 \cup A_2| \\ &= \frac{1}{36}(|A_1| + |A_2| - |A_1 \cap A_2|) \\ &= \frac{1}{36}(6 + 6 - 1) = \frac{11}{36}. \end{aligned}$$

Observe from a Venn diagram that $|A \cup B| = |A| + |B| - |A \cap B|$, and more generally that

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

This is called the *principle of inclusion and exclusion*. We will see more about it in Section 8. But for now, it is useful in the following problems.

Problem 3.8. You choose 2 Jellybellies from a cup of 30 different Jellybellies. What is the probability of choosing

a) banana and lemon

b) neither banana or lemon

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c) banana but not lemon

Problem 3.9. In a group of 80 students

- 45 take Math
- 48 take CompSci
- 38 take Stats
- 30 take Math and CompSci
- 30 take CompSci and Stats
- 23 take Math and Stats
- 15 take all three

a) How many students take none of Math CompSci or Stats?

b) Find the probability that a randomly chosen student takes Math.

c) Find the probability that a randomly chosen Stats student also takes CompSci.

Problems from the Text

Sect. 3.4: 1, 3, 4, 7, 9, 11, 14, 15

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4 Induction and the Integers

In this chapter we cover Sections 4.1 and 4.2, and very select material from the rest of Chapter 4 of the text. Also, we add some material about the 'modulo' function from Chapter 14.

4.1 Induction

A *proof by induction* proceeds as follows. For a open statement $S(n)$ where the universe for n is the set of positive integers, we can prove that $S(n)$ is true for all n by proving the following two statements.

i. $S(1)$ is true.

ii. For all $k \geq 2$, if $S(k-1)$ is true, then $S(k)$ is true.

Having done this, the principle of induction impies that $S(n)$ is true for all n.

Statement 1. is called the 'basis step' and 2. is called the 'induction step'. The assumption that $S(k-1)$ is true, in the induction step, is called the 'induction hypothesis'.

Example 4.1. Use induction to show that

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

for all positive integers n.

This is worked Example 4.1 of the text.

In practice, the universe for n may vary. But as long as there is a nice bijection between the universe of n and $\mathbb{Z}^+$, we can still do induction.

See Example 4.10 of the text, for example, where the universe of n is $\mathbb{N}$ instead of $\mathbb{Z}^+$. The principle of induction still applies here, indeed, we could replace every occurence of n with $n-1$ and 'shift' our argument to the universe $\mathbb{Z}^+$.

Strong induction

Sometime it is useful strenghen our induction assumption, and assume that a statement $S(n)$ has been proved for more values of n.

Example 4.2. Show that for all $n \geq 14$, n can be written as a sum of '3's and '8's.

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See Example 4.13 of the text

Problems from the Text

Sect 4.1: 1,3,13,15,18,19

4.2 Recursive Definitions

We can define things using a process similar to induction. This process is called recursion, it consists of a base step, and a recursion step.

The following three definitions are by recursion.

Definition 4.3. Let the notation $n!$ be defined as follows.

- Base Step: $0! = 1$
- Recursion Step: For $n \geq 1$, $n! = (n-1)! \cdot n$.

Definition 4.4. Let the general union $A_1 \cup A_2 \cup \cdots \cup A_n$ be defined for all $n \geq 2$ as follows.

- $A_1 \cup A_2 = \{x \mid x \in A_1 \vee x \in A_2\}$
- $A_1 \cup A_2 \cup \cdots \cup A_n = (A_1 \cup A_2 \cup \cdots \cup A_{n-1}) \cup A_n$

In any course in which recursion is defined, one must have the following example.

Definition 4.5. The nth Fibonacci number F_n is defined for all $n \geq 0$ as follows.

- $F_0 = 1$ and $F_1 = 1$.
- For $i \geq 2$, $F_n = F_{n-2} + F_{n-1}$.

Recursive definitions lend to inductive proofs.

See Example 4.18 of the text, proving a general version of DeMorgan's Law

Problem 4.1. Where F_i is the ith Fibonacci number, show that the following is true for all $n \geq 0$.

$$F_0 + F_1 + \cdots + F_n = F_{n+2} - 1.$$

Recall the identity about binomial coefficients that we've proved a couple of times.

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}.$$

We can use this to recursively define the binomial coefficients as follows.

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Definition 4.6. The binomial coefficients are defined by

- $\binom{0}{0} = 1$
- $\binom{n}{r} = 0$ if $r > n$ or $r < 0$
- $\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$ otherwise

Problem 4.2. Use the recursive definition of the binomial coefficients to calculate $\binom{4}{2} = 6$.

Problems from the Text

Sect 4.2: 1c,7,11,13

4.3 Integers in other bases

Integers in other bases

Definition 4.7. For integer $b \geq 2$ and $d_0, \ldots d_n \in \{0, 1, \ldots b-1\}$, the *base b integer* $(d_n d_{n-1} \ldots d_1 d_0)_b$ represents the the integer

$$d_n b^n + d_{n-1} b^{n-1} + \ldots d_2 b^2 + d_1 b + d_0.$$

So, for example, the number 99, which is represented in base 10, can be represented in base 7 as $(201)_7$, or in base 2 as $(1100011)_2$. In base 16, we need extra digits. It is standard that $(A)_{16}, (B)_{16}, \ldots, (F)_{16}$ are used for $10, 11, \ldots, 15$ respectively. Thus $90 = (5A)_{16}$.

The division algortihm (see Chapter 4 of text) ensures that any integer has a unique base b representation for any $b \geq 2$. Further, it suggests how to find it.

Example 4.8. To represent 131 in base 7 we repeatedly apply the division algorithm by dividing 7 into 131:

$$\begin{aligned} 131 = 18(7) &\quad + \quad 5 \\ 18 = 2(7) &\quad + \quad 4 \\ 2 = 0(7) &\quad + \quad 2 \end{aligned}$$

Reading up the column on the right, we get that $(245)_7 = 131$. Indeed $131 = 7(18)+5 = 7\,(7(2)+4)+5$, which gives that $131 = 2 \cdot 7^2 + 4 \cdot 7 + 5 = (245)_7$.

Problems from the Text

Section 4.3: 14,16,17

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4.4 Integers Modulo m (or clock numbers)

The following material is similar to that covered in Section 14.3 of the text, but fundamentally different. We are not defining a new number system here, simply a function of the integers.

For integers a and m, the division algorithm gives us unique integers q and r such that $a = qm + r$ with $0 \leq r \leq m - 1$. There is a lot that we can do only with the value r, throwing away the q. We denote this r by $a \mod m$, saying 'a is equal to r modulo m We write $a \equiv_m b$ to mean $(a \mod m) = (b \mod m)$. Equivalently,

Definition 4.9. For integers a, b and m, we write

$$a \equiv_m b$$

to mean that $m \mid a - b$. We say 'a is congruent to b modulo m'.

(Recall that $m \mid a - b$ means that m *divides* $a - b$; that is, there is some integer q such that $mq = a - b$.)

For example $7 \mod 4 = 3$, and $-9 \equiv_{10} 1 \equiv_{10} 11$.

The following alternate definition of $a \equiv_m b$ should be clear.

Fact 4.10. *We have $a \equiv_m b$ if and only if there is some integer k such that $a = b + mk$.*

Problem 4.3. Show that $4^n \equiv_9 3n + 1$ for all $n \geq 1$.

The following rules would simplify the above proof.

Theorem 4.11. *If $a \equiv_m b$ and $c \equiv_m d$, then*

i) $a + c \equiv_m b + d$

ii) $a - c \equiv_m b - d$

iii) $ac \equiv bd$

Proof. We do only ii). The other proofs are similar. As $a \equiv_m b$ and $c \equiv_m d$, we have by definition that $m \mid a - b$ and $m \mid c - d$. So $m \mid (a - b) - (c - d) = (a - c) - (b - d)$. By definition we get ii). □

Cancellation doesn't for work in general in the integers modulo m, because for example $2(1) \equiv_6 2(4)$ but $1 \not\equiv_6 4$. But it does work sometimes.

Theorem 4.12. *If $ac \equiv_m bc$ and $\gcd(c, m) = 1$, then $a \equiv_m b$.*

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Proof. We prove the theorem for the case that m is prime. (The general case uses the unique factorisation theorem from Chap 4 of the text, which we have skipped.) If $ac \equiv_m bc$ then $m \mid ac - bc = c(a-b)$. As m is prime, it follows that $m \mid c$ or $m \mid (a-b)$ (also a result from Chapter 4 which we skipped, but easy enough to prove). If we also have that $\gcd(c, m) = 1$, then $m \nmid c$, and so $m \mid a - b$. That is $a \equiv_m b$. $\square$

Problem 4.4. Show that 3 divide n if and only if 3 divides the sum of the digits of n.

Problem 4.5. Find the last digit of 7^{100}.

Problem 4.6. Find $15(17) - 19 \mod 7$.

Problems from the Text

Section 14.3: 1,2,16,17,31

4.5 Gray Codes

Say we want to write a computer program that counts all of the complete subgraphs of a given graph G. One way to do it would be to check for every set $S \in \mathscr{P}(V(G))$ whether or not the *subgraph induced by* S: the graph $G|_S$ with vertices S and edges

$$\{uv \in G \mid u, v \in S\},$$

is a complete graph.

To do this we have to order the subsets of $V(G)$ so that the computer can check them all.

One simple way to order all 2^n subsets of the set $[n] = \{1, \ldots, n\}$ is to associate each subset $S \subset [n]$ with its *characteristic binary string* $\chi(S) = (b_1, b_2, \ldots, b_n) \in \{0,1\}^n$, where $b_i = 1$ if $i \in S$ and $b_i = 0$ if $i \notin S$. This is clearly a bijection, and its inverse takes the binary string $b = (b_1, b_2, \ldots, b_n)$ to the set $S_b = \{i \mid b_i = 1\} \subset [n]$. Now interpreting each binary string as the binary representation of a natural number, we have an enumeration of the subsets of $[n]$: the ith subset is the set $S_{b(i)}$ where $b(i)$ is the binary representation of the integer i.

Example 4.13. According to the above discussion, the subsets of $[3] = \{1, 2, 3\}$ are ordered as follows.

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i	binary representation	subset of V
0	000	$\emptyset$
1	001	$\{1\}$
2	010	$\{2\}$
3	011	$\{1,2\}$
4	100	$\{3\}$
5	101	$\{1,3\}$
6	110	$\{2,3\}$
7	111	$\{1,2,3\}$

This is quick and easy. But for the problem of checking if each subset of vertices of a graph induces a clique, this might not be a good ordering. Consecutive subsets are completely different. It was asked, 'Can we list the subsets of $[n]$ so that consecutive subsets differ by at most one element?'

A *Gray Code* of order n is an ordering of the n element binary strings $\{0,1\}^n$ such that consecutive strings differ in at most one position. We define *a particular* Gray Code

$$G_n = (G_n(0), G_n(1), \ldots, G_n(2^n - 1))$$

of order n, recursively. (There are many different Gray Codes of order n. This is just one of them.)

For a length n binary string v, and $x \in 0, 1$, let $x|v$ denote the string we get from appending an x to the start of the string. For our base case let $G_1 = (0, 1)$, and for our recursive step, let

$$G_{n+1}(i) = \begin{cases} 0 \mid G_n(i) & \text{if } i \leq 2^n - 1 \\ 1 \mid G_n(2^{(n+1)} - 1 - i) & \text{if } i > 2^n - 1 \end{cases}$$

So $G_2 = (G_2(0), G_2(1), G_2(2), G_2(3)) = (00, 01, 10, 11)$. Viewing G_2 as this vector, we get G_3 by writing out two copies of G_2, one forwards and one backwards:

$$(00, 01, 10, 11 \mid 11, 10, 01, 00)$$

and then appending a 0 to the front of everything in the first copy, and a 1 to the front of everything in the second copy:

$$G_3 = (000, 001, 010, 011, 111, 110, 101, 100).$$

More generally, G_{n+1} is

$$G_{n+1} = (0|G_n(0), 0|G_n(1), \ldots, 0|G_n(2^n - 1), 1|G_n(2^n - 1), 1|G_n(2^n - 2), \ldots, 1|G_n(0))$$

Problem 4.7. Show that $1 \in G_n(i)$ if and only if i is congruent to 1 or 2 modulo 4.

Problem 4.8. How many Gray codes are there of order 3?

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Problem 4.9. How long is the longest path in the cube Q_n? The longest cycle? (Hint: It is not a mistake that this question is in this section. Is there a relation ship between Gray Codes and a certain path in Q_n?)

Problem 4.10. Find a Gray code of order 3 that begins with $\overline{0} = 000$ and ends with $\overline{1} = 111$. Is there a Gray code of order 2, 4 that begins with $\overline{0}$ and ends with $\overline{1}$? (Maybe this is hard, but how about order 5?)

Problem 4.11. Find $G_8(102)$. (Hint: It is a lot of work to write out the whole code. Use the recursive definition: As $102 \leq 2^7 - 1$, we have $G_8 = 0|G_7(102)$, and as $102 > 2^6 - 1 = 63$ we have $G_8 = 01|G_6(127 - 102) = 01|G_6(25)$.)

For the following problems, we view the binary string $G_n(i)$ in the Gray code as the subset $S_{G_n(i)}$ that it corresponds via the notation at the beginning of the seciton. So consecutive sets $G_n(i)$ and $G_n(i+1)$ differ by exactly one element.

Problem 4.12. Give a Gray code of order 3 in which the first set is $\{1, 3\}$.

Problem 4.13. Find all i such that $5 \in G_8(i)$.

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5 More on Counting (Relations and Functions)

Note

All of this but 5.7 is really better done in permutations and combinations.

We do material from 5.2, 5.3, 5.5, and 5.7. We recall some definitions from the other sections that are assumed to be known.

5.2 Counting by Bijections (Functions: Plain and One-to-One)

We are good at counting permutations and combinations of various types. It makes sense to relate other counting problems to these problems. We saw this when counting the number of positive integer solutions to the equation $x_1 + x_2 + x_3 = 10$, but it is a fundamental idea.

Recall that a function of sets $f : A \to B$ is

- *injective* or *one-to-one* if $f(a) = f(a') \Rightarrow a = a'$,
- *surjective* if $b \in B \Rightarrow \exists a[f(a) = b]$, and
- *bijective* it is injective and surjective.

Problem 5.1. How many functions are there from A to B? How many are injective? How many are bijective?

Given two sets A and B, and a function $f : A \to B$,

- if f is injective then $|A| \leq |B|$,
- if f is surjective then $|A| \geq |B|$, and
- if f is bijective then $|A| = |B|$.

A function $f : A \to B$ is bijective if and only if it has an inverse $f^{-1} : B \to A$, such that $f \circ f^{-1}$ and $f^{-1} \circ f$ are identity functions.

Even when we know the cardinality of a set B, it can be useful to get a bijection with other well known sets A.

Example 5.1. For example, when we were constructing Gray Codes, we used the bijection ψ from the powerset of $[n]$ to the set of binary strings of length n, defined as follows.

$$\psi : \mathscr{P}([n]) \to \{0,1\}^n : S \mapsto x_n x_{n-1} \ldots x_1,$$

where $x_i = 1$ if $i \in S$ and $x_i = 0$ otherwise.

We didn't verify that ψ was a bijection, we just believed it. To verify it, we exhibit an *inverse* function $g : \{0,1\}^n \to \mathscr{P}([n])$. Let $g(x_n x_{n-1} \ldots x_1)$ be the set that contains i if and only if $x_i = 1$.

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Counting by bijections

A *composition* of the number n is an expression of n as a sum of positive integers. So $5 = 1+1+1+1+1$, $5 = 1+2+1+1$, $5 = 2+1+1+1$, $5 = 4+1$.

Problem 5.2. How many compositions are there of the number n?

We could count the number of compositons with i summands for each i, and then add them up. (See Section 1.4 of the text.) But we will count the set by bijections.

Indeed, we will show that there are 2^{n-1} compositions of n by giving a bijection between the set of compositions, and $\mathscr{P}([n-1]$.

Start with the string $(1+1+\cdots+1+1)$. There are $n-1$ occurences of the symbol '+'. For a set $S \in \mathscr{P}([n-1])$, let $f(S)$ be the composition constructed as follows. If i is in S replace the ith occurence of '+' with ')+('. So the subset $1,3,4 \subset [8]$ corresponds to the string

$$(1)+(1+1)+(1)+(1+1+1+1).$$

Then collapse the bracketed sums: $1+2+1+4$.

To check that this is a bijection, we observe that the following function g from the set of compositions of n to $\mathscr{P}([n-1])$, is the inverse of f. For a compositon $C : a_1 + a_2 + \cdots + a_d$ of n let T be the string

$$\underbrace{(1+\cdots+1)}_{a_1 \text{ ones}} + \underbrace{(1+\cdots+1)}_{a_2 \text{ ones}} + \cdots + \underbrace{(1+\cdots+1)}_{a_d \text{ ones}}.$$

Let $g(C)$ be the set $\{a_1, a_1+a_2, \ldots, (a_1+a_2+\cdots+a_{d-1})\}$ of ordinals designating which '+'s have) and (around them.

Problem 5.3. Using the ordering of $\mathscr{P}([7])$ which you get from the bijection of it to the 7-digit binary strings (we saw this in the section on Gray Codes) , give the first three compositions of [8].

5.3 Counting by k - to - 1 functions (Onto Functions)

Generalising the idea of counting by bijections or 1-to-1 functions, we can count by k-to-1 functions.

Given two sets A and B, if we can define a function $f : A \to B$ such that for each b in B, there are k elements a of A with $f(a) = b$, then $|A| = k|B|$.

We saw this in counting the number of combinations of a set, though we didn't explicitly talk about a function.

Example 5.2. To count the number of k-combinations of n we define a function

$$f : (x_1, x_2, \ldots, x_k) \mapsto \{x_1, x_2, \ldots, x_k\}$$

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from the set of k-permutations to the set of k-combinations. For each k-combination C, there are $k!$ different k-permutations σ with $f(\sigma) = C$. So f is $k!$-to-1.

We concluded that the number $C(n,k)$ of k-combinations was

$$C(n,k) = \frac{P(n,k)}{k!} = \frac{n!}{k! \cdots (n-k)!}.$$

Problem 5.4. How many ways are there to distribute n indistinguishable items among k indistiguishable bins?

Problem 5.5. How many ways are there to distribute n indistinguishable items among k indistiguishable bins, so that no bin is empty.

Problem 5.6. How many ways are there to distribute n distinguishable items among k indistiguishable bins?

Problems from the Text

Sect 5.3: 1

5.5 The Pigeonhole Principle

A variation on the idea of counting by bijections it the idea that if we can define an onto function f from A to B, that is not a bijection, then $|A| > |B|$.

It is usually stated in the converse, with the following example:

> If there are $n+1$ pigeons in n pigeonholes, then some hole contains more than one pigeon.

This is a fairly obvious example, but the pigeonhole priciple can be used in more complicated problems.

Problem 5.7. Show that for any 5 points in the equilateral triangle of side length 1, there must be two whose distance apart is at most $1/2$.

Problem 5.8. A golfer plays 132 games over 11 weeks, playing at least one game a day. Show that there is some stretch of consecutive days in which she plays exactly 21 games.

See Example 5.48 of the text.

Problem 5.9. The *Ramsey number* $R(b,r)$ is the mininum n such that when you colour the edges of K_n, each with red or blue, there is a copy of K_b with all edges blue or a copy of K_r with all edges red. Find $R(3,3)$. Show that $8 \leq R(3,4) \leq 10$. (Can you show that $R(3,4) = 9$)?)

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Example 5.3. Consider any two colouring of the grid points of a 3×9 grid. Show that there must be a rectangle all of whose corners get the same colour.

Problem 5.10. Find the minimum n such that any two colouring of the grid points of a $3 \times n$ grid has a rectangle all of whose corners get the same colour. (How about $4 \times n$?)

Problem 5.11. 1502 people are seated in a row of 2002 chairs. Show that there are 3 consecutive non-empty chairs.

Problems from the Text

Sect 5.5: 7,9,11,12,23

5.7 Computational Complexity

This chapter is Section 5.7 of the text

We look at how a function $f : \mathbb{Z} \to \mathbb{R}$ grows when $n \in \mathbb{Z}$ is large- what the graph of f looks like from far away.

Example 5.4. We would like so say approximately that

- $2n^2 - 3n + 7$ grows like n^2,
- $100\sqrt{n} + \ln n$ grows like $\sqrt{n}$, and
- $2^n - n!$ grows like $n!$.

This will be the statement that f is *of order* g. We start with a one sided version of this.

Definition 5.5. Let $f, g : \mathbb{Z} \to \mathbb{R}$ be functions. We write $f = O(g)$, and say that 'f is Big-oh of g' or 'f is dominated by g' or 'f is of order at most g' if there exists an integer k and a real number m such that for all $n \geq k$,

$$|f(n)| \leq m \cdot |g(n)|.$$

Hint

See Example 5.65 of text.

Problem 5.12. Show that $2n^2 - 3n + 7$ is $O(n^2)$.

Usually big-oh notation is used to simplify the function that we write, but we can also show the following.

Problem 5.13. Show that $n^2 = O(2n^2 - 3n + 7)$.

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Solution

For $n \geq 0$ we have that

$$|n^2| = n^2 \leq 2n^2 - 3n + 7 = |2n^2 - 3n + 7|$$

if and only if $0 \leq n^2 - 3n + 7$. This is true for $n^2 > 3n$ so if $n > 3$.
Thus for all $n \geq 3$ we have that $|n^2| \leq |2n^2 - 3n + 7|$, and so $n^2 = O(2n^2 - 3n + 7)$

But big-oh notation is far from symmetric. Clearly $n = O(n^2)$ while $n^2 \neq O(n)$. Oh, is this clear. How would we show that $f \neq O(g)$.

Problem 5.14. Show that $2n^2 - 3n + 7 \notin O(n)$.

Solution

If $2n^2 - 3n + 7 \in O(n)$, then there are m, k such that if $n > k$ then $|2n^2 - 3n + 7| < m|n|$. But

$$\begin{aligned} & |2n^2 - 3n + 7| < m|n| \\ \iff \quad & 2n^2 - 3n + 7 < mn \qquad \forall n \geq 10 \\ \iff \quad & 2n^2 + 7 < (m+3)n \end{aligned}$$

This is not true for $n \geq m + 3$. So $|2n^2 - 3n + 7| < m|n|$ is not true for $n \geq \max(m + 3, 10)$.

Problem 5.15. Show that 2^n is $O(n!)$.

Hint

See Example 5.66 of text

Problem 5.16. Show that $2^n - n!$ is not $O(2^n)$.

These proofs should feel a lot like 'limits' from calculus. In fact, the definition of 'f is $O(g)$' can be written that there is $m \geq 0$ such that $|f(n)|/|g(n)| \leq m$ for large enough n. As such, a quick way to show that f is dominated by g is to show that

$$\lim_{n\to\infty} \frac{|f(n)|}{|g(n)|} \neq \infty.$$

Example 5.6. We show that $100\sqrt{n} + \ln n \in O(\sqrt{n})$. Indeed,

$$\begin{aligned} \lim_{n\to\infty} \frac{|100\sqrt{n} + \ln n|}{|\sqrt{n}|} &= \lim_{n\to\infty} \frac{100\sqrt{n} + \ln n}{\sqrt{n}} \\ &= \lim_{n\to\infty} 100 + \frac{\ln n}{\sqrt{n}} \\ &\overset{l'hopital}{=} \lim_{n\to\infty} 100 + \lim_{n\to\infty} \frac{1/n}{1/2n^{-1/2}} \\ &= 100 + \lim_{n\to\infty} \frac{2}{n} = 100 \end{aligned}$$

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So $\lim_{n\to\infty} \frac{|100\sqrt{n}+\ln n|}{|\sqrt{n}|} = 100$ so $|100\sqrt{n} + \ln n| < 101|\sqrt{n}|$ for large enough n.

This is no use, however in showing that f is not in $O(g)$:

$$f(n) = \begin{cases} 1 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

is $O(1)$ but $\lim_{n\to\infty} \frac{|f(n)|}{1}$ does not exist.

Problem 5.17. Show that $f = O(g)$ if and only if there are constants k and $m > 0$, and a function h such that $h(n) \geq f(n)$ for all $n > k$, and

$$\lim_{n\to\infty} \frac{|h(n)|}{|g(n)|} = m.$$

5.7.1 Lower Bounds

Definition 5.7. For functions $f, g : \mathbb{Z} \to \mathbb{R}$, f is $\Omega(g)$, read 'big omega', if there exist m, k such that $\forall n \geq k$

$$|f(n)| \geq m|g(n)|.$$

Alternately: f is $\Omega(g)$ if and only if g is $O(f)$. Indeed $|f| \geq m|g|$ if and only if $|g| \leq 1/m|f|$.

Definition 5.8. If f is $O(g)$ and $\Omega(g)$ then it is $\Theta(g)$, read 'big theta of g'. Equivalently, f is $\Theta(g)$ if there are k, m_1, m_2 such that for all $n \geq m$

$$m_1|g(n)| \leq |f(n)| \leq m_2|g(n)|.$$

Equivalently, f is $\Theta(g)$ if and only if f is $O(g)$ and g is $O(f)$; we say f and g have the same order.

Problem 5.18. Show that $f(n) = 1^2 + 2^2 + 3^2 + \ldots n^2$ is $\Theta(n^3)$.

From the examples/problems of this section, we can infer a heirarchy of orders:

$$\Theta(1) < \Theta(\log n) < \Theta(n^{.02}) < \Theta(n) < \Theta(n^5) < \Theta(C^n) < \Theta(n!)$$

(where $C > 1$).

Problems from the Text

Sect 5.7: 1, 3, 12, 16

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5.7.2 Extra: Complexity of a Problem

In complexity theory, a *problem* is something such as

- PRIMES: Decide for a given integer a if a is prime.
- CYCLIC: Decide for a given graph G if G contains a cycle.

In these problems, a particular integer a or graph G is called an *instance* of the problem. An *algorithm* for a problem is a set of steps one can perform to solve the problem.

Note

Usually an algorithm terminates by returning 'Yes'. If it terminates without returning 'Yes', we say it returns 'No'. So we don't need to say the bracketed things in the above algorithm. If it doesn't terminate, we won't consider it. This is an issue called *computablity*, which we don't talk about.

Example 5.9. The following is an algorithm for the PRIMES problem.

Input: An integer a.

Output: 'Yes' if a is prime. ('No' otherwise.)

Algorithm: For integer i from 2 to $\sqrt{n}$ check if a/n is integer. (If it is, return 'Yes' and quit. If 'Yes' is not returned for any i, return 'No'.)

The *(computational) complexity of an algorithm* for a problem is a function f such that for **any** instance of the problem of *size* x, the algorithm takes **at most** *time* $f(x)$.

How we measure size and time depends on the problem. For PRIMES size would either be a itself, or the number of digits, or binary digits, of a. Time might be the time it takes a particular computer to run the algorithm, but is usually counted as the number of basic operations: sum or product of a binary digit. Certainly the *unit* of size and time can effect the complexity, so one must be careful about what they are. We won't look at the complexity of the above algorithm, the issue of size and time units adds complications that we don't want to think about now.

Graphs problems are a bit easier. We usually let size be the number of vertices, and time be the number of times we check if two vertices are adjacent.

Here is a simple minded algorithm for CYCLIC.

INPUT: A graph G.

OUTPUT: 'Yes' if G contains a cycle.

Algorithm: For every ordered subset $S = (v_1, v_2 \ldots, v_d)$ of vertices of G, return 'Yes' if $v_i \sim v_{i+1}$ (indices modulo d) for all $i \in [d]$.

For a graph on n vertices, there are

$$\sum_{i=1}^{n} \frac{n!}{i!} = O((n+1)!)$$

ordered subsets of vertices and there are $O(n)$ adjacencies to check for each, so this algorithm for CYCLIC has complexity $O((n+2)!)$.

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Here is a better algorithm:

We have two sets, think of A as the 'checked vertices' B as the other vertices. We will check each vertex, by which we mean, we will check if it is in a cycle with other checked vertices. (We assume that G is connected. If not, we can run this algorithm on each component.)

i. Set $A_1 = \{v_1\}$ and $B_1 = V(G) - A_1$ for some vertex v_1 of G.

ii. For $i = 1 \dots n$ do the following.

   (a) Select a vertex b in B_i that is adjacent to some vertex in A_i.

   (b) If b is adjacent to more than one vertex in A_i, return 'Yes'; otherwise set $A_{i+1} = A_i \cup \{b\}$ and $B_{i+1} = B_i - \{b\}$.

Problem 5.19. (Not testable.) Show that this algorithm will return 'Yes' if and only if G contains a cycle.

The outer loop happens at most n times and in the inner loop, we have to check at most n adjacencies. So this algoritm has complexity $O(n^2)$.

The *(computational) complexity of a problem* is the complexity of the fastest algorithm for the problem. We see that CYCLIC has complexity $O(n^2)$.

Problem 5.20. (Not testable.) Can you improve this?

For the problem MAX-CLIQUE of finding the largest complete subgraph in a given graph G, the best known algorithm we have is $\Theta(c^n)$ for some c. To decide if there is a cycle, there is an algorithm of order $O(n^2)$. The first one is *exponential time*, we consider this a difficult problem, the second is *polynomial time*, we consider it easy. For combinatorial problems we usually say it is *easy* or *hard* depending on whether or not there is a polynomial time algorithm for solving it.

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8 Principle of Inclusion and Exclusion

This problem is a little hard right now.

Problem 8.1. How many ways are there to distribute n distinguishable items among k distiguishable bins, such that no bin is empty? What if the bins are indistiguishable.

Here are the tools to do it.

Inclusion and Exclusion

We observed earlier, by looking at the Venn Diagram, that for sets A and B,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

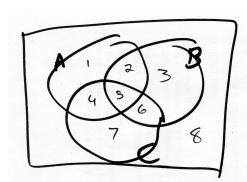
We can use this to count the number of integers in [60] that are multiples of 2 or 3:

Solution

Let A be the set of multiples of 2 and B be the set of multiples of 3. Then $|A| = 30$ and $|B| = 20$ and $A \cap B$ is the set of multiples of 6, so $|A \cap B| = 10$. So by the above identity we have that

$$|A \cup B| = |A| + |B| - |A \cap B| = 30 + 20 - 10 = 40$$

Problem 8.2. How many arrangements of [60] start or end with an odd digit?



Looking at the Venn Diagram for three sets, we see that $|A \cup B \cup C|$ is equal to

$$\begin{aligned} & |A| + |B| + |C| \\ - \quad & |A \cap B| - |A \cap C| - |B \cap C| \\ + \quad & |A \cap B \cap C| \end{aligned}$$

Clearly this generalises to more sets. We introduce some notation to simplify the statement. Let S be a set and for $i \in [n]$ let c_i be a property that can be

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satisified by elements of S. Then we let S_i be the subset of S of elements that satisfy c_i.

Example 8.1. Let's count the number of arrangements of the letters of the word CHEMIST that have H before T, or T before E, or E before M.

Let A be the set of arrangements of the letters. Let c_1 be the property of an arrangement that H comes before T. Let c_2 be the property of an arrangement that T comes before E. Let c_3 be the property of an arrangement that E comes before M.

So A_1 is the set of arrangements with H before T, etc.

Then $|A_1| = \frac{7!}{2!} = |A_2| = |A_3|$, $|A_1 \cap A_2| = \frac{7!}{3!} = |A_2 \cap A_3|$, $|A_1 \cap A_3| = \frac{7!}{2!2!}$, and $|A_1 \cap A_2 \cap A_3| = \frac{7!}{4!}$.

Using the above identity, we can put this together to get $|A_1 \cup A_2 \cup A_3| = 7!(\frac{3}{2!} - (\frac{2}{3!} + \frac{1}{2!2!}) + \frac{1}{4!}$.

Let's introduce a little more notation. Let $S_1, \ldots, S_n$ be subsets of a set S. Then for any set $X \subset [n]$ of subset indices, let

$$N(X) = |\cap_{i \in X} S_i|.$$

We write $N(1,2,3)$ for $N(\{1,2,3\})$ to make things cleaner.

Example 8.2. In the Venn Diagram above let $A = S_1, B = S_2$, and $C = S_3$. Then the equation given above can be written

$$\begin{aligned} |S_1 \cup S_2 \cup S_3| &= N(1) + N(2) + N(3) - (N(1,2) + N(2,3) + N(1,3)) + N(1,2,3) \\ &= \sum_{i=1}^{3} (-1)^{i+1} \sum_{X \in \binom{[3]}{i}} N(X) \end{aligned}$$

(Recall that $\binom{S}{i}$ is the set of i element subsets of S.)

Theorem 8.3 (Principle of Inclusion and Exclusion)**.** *With notation as above,*

$$|S_1 \cup S_2 \cup \cdots \cup S_d| = \sum_{i=1}^{d} (-1)^{i+1} \sum_{X \in \binom{[d]}{i}} N(X).$$

Problem 8.3. Show that the following is equivalent to the Principle of Inclustion and Exclusion. Where each S_i is a subset of a universe S, and $N(\emptyset) = |S|$,

$$|\overline{S_1 \cup S_2 \cup \cdots \cup S_n}| = \sum_{i=0}^{d} (-1)^{i} \sum_{X \in \binom{[d]}{i}} N(X).$$

Often, this is called the Dual form of the Principle of Inclustion and Exclusion.

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In Problem 5.1 we forgot to ask how many surjective functions there are from A to B. No we didn't we left it out because it is a little harder. But now we should be able to do it.

Problem 8.4. If $m = |A| \geq |B| = n$ show that the number of surjective functions from A to B is

$$\sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)^m.$$

Hint: Let S be the set of functions from A to $B = \{b_1, \ldots, b_n\}$, and for a function $f \in S$, let c_i be the property that b_i is not in the range of f. What set describes the surjective functions? How big is it?

Now we should be able to answer Problem 8.1.

Problems from the Text

Sect 8.1: 2, 5, 11, 17

8.3 Derangements

In a permutation of $[n]$ we say that an element i is *fixed* if it is the ith element. In the permutation $(2, 4, 3, 1)$ only the element 3 is fixed. A *derangement* of $[n]$ is a permutation in which no elements are fixed. Let $D(n)$ be the number of derangements of $[n]$. There are $D(3) = 2$ derangements of $[3]$: $(2, 3, 1)$ and $(3, 1, 2)$. There are $D(4) = 9$ derangements of $[4]$. What is $D(n)$?

Where S is the set of permutations of $[n]$ and c_i is the property of a permutation that it fixes i, we want to count $D(n) = |\overline{S_1 \cup S_2 \cup \cdots \cup S_n}|$. For this, we use the Dual form of the Principle of Inclusion and Exclusion.

We have that $N(1) = (n-1)!$ and the same for all c_i. Further $N(1, 2, \ldots, i) = (n-i)!$, and more generally $N(X) = (n-i)!$ for any $X \in \binom{[n]}{i}$. So the number of derangements of $[n]$ is

$$\begin{aligned}
D(n) &= \sum_{i=0}^{n} (-1)^i \sum_{X \in \binom{[n]}{i}} N(X) \\
&= \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)! \\
&= n! - n(n-1)! + \binom{n}{2}(n-2)! - \binom{n}{3}(n-3)! + \ldots \\
&= n! \left(1 - \binom{n}{1}\frac{(n-1)!}{n!} + \binom{n}{2}\frac{(n-2)!}{n!} - \ldots \binom{n}{n}\frac{0!}{n!}\right) \\
&= n!(1 - 1 + (1/2!) - (1/3!) + (1/4!) - \cdots \pm (1/n!))
\end{aligned}$$

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Now from calculus you might recall the Taylor expansion of e^x about 0:

$$e^x = e^0 + \frac{e^0}{1!}x + \frac{e^0}{2!}x^2 + \dots$$

Putting $x = -1$ we get that $D(n) \approx \frac{n!}{e}$ is a pretty good approximation for $D(n)$.

Note

The error $\varepsilon = |D(n) - \frac{n!}{e}|$ is

$$\begin{aligned} \varepsilon &= n!\,(1/(n+1)! - 1/(n+2)! + \dots) \\ &= n!\,((n+1/n+2)(1/(n+1)!) + (n+3/n+4)(1/(n+3)!) + \dots) \end{aligned}$$

Which is greater than $n!\,(1/(n+1)! + 1/(n+3)! + \dots) > n! \cdot 1/(n+1)!(n^2/n^2 - 1) \approx n$. But it is quite small compared to $D(n)$ itself. (How small?)

Problems from the Text

Sect 8.3: 6,9,13

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9 Generating Functions

9.1 Introduction

Consider the following problem:

Example 9.1. How many ways can we give 10 dimes to 3 children (Archie, Betty, and Jughead), such that

i. each child gets at least one dime,

ii. Archie gets at most four dimes,

iii. Betty gets at least three dimes, and

iv. Jughead gets an even number of dimes.

We can certainly deal with the first three restrictions, but the fourth gets tricky. And what if we add more? At a certain point, it seems easiest to list the possible distributions of dimes, and find there are 8.

Archie	Betty	Jughead
1	3	6
1	5	4
1	7	2
2	4	4
2	6	2
3	3	4
3	5	2
4	4	2

Of course a listing like this is unsatisfactory. Let's try something else. The possible distribution of dimes to Archie is 1 or 2 or 3 or 4. Let's encode this into the following polynomial

$$f_A = x + x^2 + x^3 + x^4.$$

We do the same with the other children, getting $f_B = x^3 + x^4 + \ldots x^{10}$, and $f_J = x^2 + x^4 + x^6 + x^8 + x^{10}$. (Perhaps you wanted that $f_J = x^2 + x^4 + x^6$. Fine. But it doesn't matter.)

Now consider the co-efficient of x^{10} in the polynomial

$$f = f_A \cdot f_B \cdot f_J = (x + x^2 + x^3 + x^4)(x^3 + x^4 + \ldots x^{10})(x^2 + x^4 + x^6).$$

We get one term by picking x from f_A and x^3 from f_B and x^6 from f_J. And we get another by by picking x from f_A and x^5 from f_B and x^4 from f_J. And...

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See what's happening? Look at the rows of the table above. The co-efficient of x^{10} in f is counting the number of ways we can distribute the 10 dimes.

'So?' you say. 'I still have to compute that co-efficient. It is the same work.'

Not always, you doubting Nancy.

If we are interested in distributing n dimes instead of 10 then we are going to need to alter f_B and f_J.

$$f_B = x^3 + x^4 + x^5 + \cdots = x^3(1 + x + x^2 + x^3 + \dots) = \frac{x^3}{1-x}, \text{ and}$$

$$f_J = x^2 + x^4 + x^6 + x^8 + \cdots = x^2(1 + x^2 + x^4 + \dots) = \frac{x^2}{1-x^2}.$$

Here I've used identities for the geometric series which we will discuss soon. We can also rewrite f_A as

$$f_A = x + x^2 + x^3 + x^4 = x(1 + x + x^2 + x^3) = \frac{x(1-x^4)}{(1-x)}.$$

So

$$f = x^6 \frac{1-x^4}{(1-x^2)(1-x)^2} = x^6 \frac{1+x^2}{(1-x)^2}.$$

The co-efficient of x^{10} in this is the co-efficient of $x^{10-6} = x^4$ in $(1+x^2)(1-x)^{-2}$, is the coefficient of x^4 in $(1-x)^{-2}$ plus the coefficient of $x^{4-2} = x^2$ in $(1-x)^{-2}$.

Now believing for the moment that

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

we get that the 2nd coefficient is 3 and the 4th is 5. So there are $3+5=8$ ways to distribute 10 dimes, and $(n-7)+(n-5) = 2n-12$ ways to distribute n dimes.

Now that is some pretty neat math. We encoded our problem as a polynomial, and the coefficients of the polynomial gives us the answer to the problem for different values of n. And wha'ts more, using some identities that we must recall about power series, we can find this coefficients in a nice (?) way.

Let's look at it a bit closer.

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9.2 Definitions and Calculation Techniques

For a sequence $\{a_k\}_{k=0}^{\infty} = a_0, a_1, a_2, \dots$ of real numbers, the function

$$f(x) = \sum_{k=0}^{\infty} a_i x^i$$

is called the *generating function for* $\{a_i\}$.

Note

We are not interested is such things as convergence of the function; it is an algebraic construction. For all of our purposes we may always restrict ourselves to partial sum of a finite number of terms, and the we get convergence for free.

9.2.1 Some Basic Generating Functions

Some generating functions that we can easily derive are the following.

Example 9.2. For $n \geq 0$ an integer, we have

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n.$$

This is the generating function for the sequence

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \dots, \binom{n}{n}, 0, 0, 0.$$

Problem 9.1. The function $(1+ax)^n$ generates what sequence?

Recalling the geometric series or initial geometric series identities, we have the following.

Example 9.3. One can check that $(1-x^n) = (1-x)(1+x+x^2+\cdots+x^{n-1})$ and that $1 = (1-x)(1+x+x^2+x^3+\dots)$ by polynomial long division, so we have that

- $\frac{1-x^n}{1-x} = 1+x+x^2+\cdots+x^{n-1}$ and
- $\frac{1}{1-x} = 1+x+x^2+x^3+\dots$.

These generate the series $1, 1, 1, \dots, 1, 0, 0, \dots, 0$ and $1, 1, 1, \dots$ respectively.

Example 9.4. Using the above, we get that $\frac{x^2(1-x^7)}{1-x}$ generates $0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0,$

Example 9.5. Putting $y = 2x$ into $\frac{1}{1-y}$ we get that

$$\frac{1}{1-2x} = 1 + 2x + 2^2x^2 + 2^3x^3 + \dots$$

generates the sequence $a_i = 2^i$.

Problem 9.2. Using the formula from Example 9.3, find the sequence generated by $\frac{1}{1-x^2}$.

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Problem 9.3. What is the generating function for $1, -1, 1, -1, 1, -1, \ldots$? How about for $-1, 1, -1, 1, -1, 1, \ldots$? Add the generating functions for $1, -1, 1, -1, \ldots$ to the one for $1, 1, 1, 1, \ldots$ to get one for $2, 0, 2, 0, 2, 0, \ldots$. Compare this with Problem 9.2.

Note

To justify taking the derivative on the right hand side term-by-term, we could use Weierstrauss, and observe that the resulting series converges uniformly on some interval around 0, but we rather just ignore this, observing that it is valid for any partial sum of the series.

Example 9.6. Taking the derivative of both sides of the the identity $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots$, we get the idenity we used in Example 9.1:

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \ldots.$$

This is the generating function for $a_i = i + 1$, so $x/(1-x)^2$ is the generating function for $a_i = i$.

Problem 9.4. Taking another derivative of the identity in Example 9.6, find a generating function for the series $a_i = i^2$.

Notice that the identity from Example 9.6 can be written as

$$(1-x)^{-2} = 1 + 2x + 3x^2 + \ldots.$$

We can view this as a case of the following extension of the binomial theorem.

Theorem 9.7. *Let n be* any real *number, (not just a positive integer). Then*

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k,$$

where we define $\binom{n}{k}$ for any real n and positive integer k as

$$\binom{n}{k} = \frac{(n)(n-1)\ldots(n-k+1)}{k!},$$

and $\binom{n}{0} = 1$.

To prove this, first we recall that the Maclauren expansion of a infinitely differentiable function $f(x)$ is the following infinite series

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{[k]}(0)}{k!} x^k$$

where $f^{[k]}(0)$ is the kth derivative of f, evaluated at $x = 0$.

Now we prove the theorem.

Proof. Observe first that the derivative of $f(x) = (1+x)^n$ is $f^{[1]}(x) = f'(x) = n(1+x)^{n-1}$ and the second derivative is $f^{[2]}(x) = n(n-1)(1+x)^{n-2}$. By induction one can see that

$$f^{[k]}(x) = n(n-1)(n-2)\ldots(n-k+1)(1+x)^{n-k}$$

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and so $f^{[k]}(0) = n(n-1)(n-2)\dots(n-k+1)$.

Thus the Maclauren expansion of $(1+x)^n$ is

$$(1+x)^n = \sum_{k=0}^{\infty} \frac{n(n-1)(n-2)\dots(n-k+1)}{k!} x^k = \binom{n}{k} x^k$$

as needed. □

The following is useful in computing binomial coefficients when n is negative. For $-n < 0$ we have

$$\begin{aligned} \binom{-n}{k} &= \frac{(-n)(-n-1)\dots(-n-k+1)}{k!} \\ &= (-1)^k \frac{(n+k-1)\dots(n+1)(n)}{k!} \\ &= (-1)^k \binom{n+k-1}{k}. \end{aligned}$$

Thus for $n > 0$ we have

$$\frac{1}{(1-x)^n} = (1+(-x))^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} (-1)^k x^k = \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k.$$

See Table 9.2 of the text for a list of the generating functions of this section.

Problems from the Text

Sect 9.2: 1,2,3,27

9.2.2 Encoding Problems into Generating functions

With now a feeling for some of the generating functions we can express nicely, lets try to encode some problems.

The basic idea, which we saw in Example 9.1 that the number of positive integer solutions to

$$x_1 + x_2 + x_3 = n$$

is the nth coefficient of the generating function

$$\begin{aligned} f &= f_1 \cdot f_2 \cdot f_3 \\ &= (1 + x + x^2 + \dots) \cdot (1 + x + x^2 + \dots) \cdot (1 + x + x^2 + \dots) \\ &= \left(\frac{1}{1-x}\right)^3 \end{aligned}$$

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If we add the restriction that $x_1 \leq 4$, then f_1 becomes

$$f_1(x) = 1 + x + \ldots x^4 = \frac{1 - x^5}{1 - x}.$$

Or if we add the restriction that $x_1 > 2$ then it becomes

$$f_1(x) = x^2 + x^3 + x^4 + \cdots = \frac{x^2}{1 - x}.$$

Or if we add the restriction that x_1 is even, it becomes

$$f_1(x) = 1 + x^2 + x^4 + \cdots = \frac{1}{1 - x^2}.$$

Problem 9.5. If x_1 must be odd, what is f_1?

Example 9.8. How many non-negative integer solutions are there to the equation

$$3x_1 + 2x_2 + x_4 + 2x_4 = 20?$$

The generating polynomial for possible values of x_1 is $f_1(x) = 1+x^3+x^6+\cdots = 1/(1 - x^3)$. In a similar way we get the generating polynomial for the total solution is

$$f(x) = \frac{1}{(1 - x)(1 - x^2)^2(1 - x^3)}.$$

Example 9.9. There are an unlimited number of jellybeans of each of four colours: yellow, orange, pink and green. How many ways can Douglas select n of them so that there are at most 3 yellow, 10 orange, 12 pink, and exactly as many green as there are pink plus yellow?

We imagine that Douglas doesn't choose any green jellybeans. Instead, he chooses an even number of yellow ones and an even number of pink ones and when he is done, paints half of them green. So he can choose at most 6 yellow and at msot 24 pink.

We make a polynomial for each of orange, yellow, and pink: $f_o = 1+x^2+x^4+\cdots+x^{10} = \frac{1-x^{11}}{1-x}$, $f_y = 1+x^2+x^4+x^6 = \frac{1-x^8}{1-x^2}$, and $f_p = 1+x^2+\cdots+x^{24} = \frac{1-x^{26}}{1-x^2}$.

The nth coefficient of the generating function

$$f_o \cdot f_y \cdot f_p = \frac{(1 - x^{11})(1 - x^8)(1 - x^{26})}{(1 - x)(1 - x^2)^2},$$

counts the number of solutions when Douglas selects n jellybeans. We know how to find the coefficient of $\frac{1}{1-x^2}$ or even $\frac{1}{(1-x^2)^2}$ using the generalised binomial theorem with $y = (-x^2)$. But this looks tricky.

Problems from the Text

Sect 9.1: 1,2,5

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9.2.3 Finding Coefficients

Now, given a generating function, we will want to extract the coefficient of x^n. This can take some ingenuity.

Let's see some examples.

Example 9.10. To find the coefficient of x^n in $f(x) = (1-2x)^{-7}$ we observe that

$$f(x) = (1 + (-2x))^{-7} = \sum_{k=0}^{\infty} \binom{-7}{k} (-2x)^k$$

and so the coefficient of x^n is $(-2)^n \binom{-7}{n} = (-2)^n \binom{7+n-1}{n} (-1)^n = 2^n \binom{7+n-1}{n}$.

Example 9.11. To find the coefficient of x^n in $\frac{ax^2}{(1-2x)^7}$ we observe that this is a times the coefficient of x^{n-2} in $f(x) = (1-2x)^{-7}$ from the last example. So is $a \cdot 2^{n-2} \binom{5+n-1}{n-2}$. Careful though, this only holds for $n \geq 2$. For $n = 0, 1$ the coefficient is 0.

Example 9.12. To find the coefficient of x^n in $\frac{x+x^2}{(1-2x)^7}$ we observe that this is

$$\frac{x}{(1-2x)^7} + \frac{x^2}{(1-2x)^7}$$

so using the previous example is $2^{n-1} \binom{6+n-1}{n-1} + 2^{n-2} \binom{5+n-1}{n-2}$.

It looks like we can deal with any thing of the form

$$f(x) = \frac{p(x)}{(1 + (cx))^d}$$

for a polynomial $p(x)$, although we would like the polynomial not to have too many terms.

If the bottom is more complicated, we try to factor it. If we cannot, we will have to use the method of partial fractions.

Example 9.13. To find the coefficient of x^n in $\frac{x(1-x^2)}{(1-x)(1-2x)^7}$ we observe that $\frac{1-x^2}{(1-x)(1-2x)^7} = \frac{x+x^2}{(1-2x)^7}$. and so this is the same as the previous example.

Example 9.14. To find the coefficient of x^n in $\frac{(x^2)}{(1-x)(1-2x)}$ we us the method of partial fractions. We could do this for the given function directly, but we rather use partial fractions to split up $\frac{1}{(1-x)(1-2x)}$.

We say that

$$\frac{1}{(1-x)(1-2x)} = \frac{A}{1-x} + \frac{B}{1-2x} = \frac{A(1-2x) + B(1-x)}{(1-x)(1-2x)}$$

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for some A and B and so get that $1 = A(1-2x)+B(1-x) = A+B-x(2A+B)$. Equating coefficients, this yields $1 = A+B$ and $2A+B = 0$. So $A = -1$ and $B = 2$.

We thus rewrite our generating function as

$$\frac{(x^2)}{(1-x)(1-2x)} = x^2(\frac{-1}{1-x} + \frac{2}{1-2x}) = \frac{-x^2}{1-x} + \frac{2x^2}{1-2x}.$$

The nth coefficient of the $\frac{1}{1-x}$ is 1, so of $\frac{-x^2}{1-x}$ is -1 (for $n \geq 2$.) The nth coefficient of $\frac{1}{1-2x}$ is 2^n so of $\frac{2x^2}{1-2x}$ is $2 * 2^{n-2} = 2^{n-1}$ (for $n \geq 1$).

So the 0th coefficient is 0, the 1st is $2^0 = 1$, and for $n \geq 2$ the nth is $2^{n-1} - 1$.

Example 9.15. Find the coefficient of x^n in $g(x) = \frac{1}{1-x-x^2}$.

Observe that (x^2+x-1) has roots $\frac{-1\pm\sqrt{5}}{2}$. Writing $\alpha = \frac{1+\sqrt{5}}{2}$, we have that

$$1/\alpha = \frac{2}{1+\sqrt{5}} = \frac{2(1-\sqrt{5})}{(1+\sqrt{5})(1-\sqrt{5})} = -\frac{1-\sqrt{5}}{2} = \frac{-1+\sqrt{5}}{2},$$

So these roots can be written $-\alpha$ and $1/\alpha$. This lets us factor $(1-x-x^2)$ as

$$(1-x-x^2) = -(x^2+x-1) = (\alpha+x)(1/\alpha-x) = (1+x/\alpha)(1-\alpha x).$$

We split g by partial fractions:

$$\begin{aligned} g(x) &= \frac{1}{1-x-x^2} = \frac{1}{(1+x/\alpha)(1-\alpha x)} \\ &= \frac{A}{1+x/\alpha} + \frac{B}{1-\alpha x} \\ &= \frac{A+B-x(B/\alpha - A\alpha)}{(1+x/\alpha)(1-\alpha x)}. \end{aligned}$$

Equating coefficients gives $A+B = 1$ and $B/\alpha = A\alpha$. Using that $\alpha = \frac{1+\sqrt{5}}{2}$ one can solve this and get that $B = \alpha/\sqrt{5}$ and $A = 1/(\alpha\sqrt{5})$. So

$$\begin{aligned} g(x) &= \frac{B}{1-\alpha x} + \frac{A}{1-(-1/\alpha)x} \\ &= \frac{1}{\sqrt{5}}\left(\frac{\alpha}{1-\alpha x} + \frac{1/\alpha}{1-(-1/\alpha)x}\right) \\ &= \frac{1}{\sqrt{5}}\left(\alpha(1+\alpha x+\alpha^2x^2+\dots) - (-1/\alpha)(1+(-1/\alpha)x+(-1/\alpha)^2x^2+\dots)\right) \end{aligned}$$

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The nth coefficient of which is

$$\frac{1}{\sqrt{5}}(\alpha^{n+1} - (-1/\alpha)^{n+1}) = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} + \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right).$$

We will use this later.

Problems from the Text

Sect 9.2: 4,5,6,9

9.2.4 Putting it all together

Example 9.16. How many ways can Tim give out 10 dimes to 3 children so that each child gets between 3 and 5 dimes? We need the coefficient of x^{10} in $(x^3 + x^4 + x^5)^3 = x^3\frac{(1-x^4)^3}{(1-x)^3}$ which is the coefficient of x^7 in

$$(1-x^4)^3(1-x)^{-3} = [1 - 3x^4 + 3x^8 - x^{12}][\binom{-4}{0} - \binom{-4}{1}x + \binom{-4}{2}x^2 - \dots]$$

To find the coefficient of x^7, we observe that we get contribution to it from each of the terms 1 and $3x^4$ in the first polynomial. So it is $1 \cdot \binom{-4}{8} + (-3)\binom{-4}{4} = \binom{11}{8} - 3\binom{7}{4}$.

See the following example 9.17 in the text.

Example 9.17. How many four element subsets of [20] contain no consecutive integers?

Problems from the Text

Sect 9.2: 10,11,12,15

9.4 Exponential Generating Functions

The *ordinary generating functions* we have looked at up to now are good for combination problems. We now consider *exponential generating functions* which will be good of permutation problems.

For a sequence $\{a_k\}_{k=0}^{\infty} = a_0, a_1, a_2, \dots$ of real numbers, the function

$$f(x) = \sum_{k=0}^{\infty} a_i \frac{x^i}{i!}$$

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is called the *exponential generating function for* $\{a_i\}$.

Example 9.18. We have seen that $(1+x)^n$ is the generating function for the binomial coefficients $\binom{n}{i} = C(n,i)$ which count the number of i-combinations of $[n]$. Recalling that the number of i-permutations of $[n]$ is $P(n,i) = i!C(n,i) = i!\binom{n}{i}$ we get that

$$\begin{aligned}(1+x)^n &= \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots \\ &= P(n,0) + P(n,1)x + P(n,2)\frac{x^2}{2!} + P(n,3)\frac{x^3}{3!} + \dots\end{aligned}$$

So $(1+x)^n$ is the exponential generating function for $P(n,i)$.

Example 9.19. As the Maclaurin expansion of e^x is

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

e^x is the exponential generating function for $1, 1, 1, \dots$.

Problem 9.6. What sequence is e^x the ordinary generating function for?

The following is our typical example.

Example 9.20. Find the exponential generating function for the sequence $\{a_i\}$ where a_i is the number of n-permutations of the *multiset* $\{A, A, A, B, B, C, D, E, F\}$?

Let's first solve the problem for, say, $n = 4$. There are $\frac{4!}{2!1!1!}$ 4-permutations with say, two A, a B and a D. The total number, a_4, of 4-permutations is

$$4! \sum \frac{1}{m_A! m_B! \dots m_F!},$$

where we sum over all choices of non-negative integer solutions to $m_A + \dots + m_F = 4$ with $m_A \le 3$, $m_B \le 2$, etc.

Now consider the x^4 coefficient in

$$f(x) = (1 + x + \frac{x^2}{2!} + \frac{x^3}{3!})(1 + x + \frac{x^2}{2!})(1+x)(1+x)(1+x)(1+x).$$

When we pick the x^{m_A} term from the first polynomial, the x^{m_B} term from the second, etc, we get a contribution to the x^4 coefficient of $f(x)$ if and only if the m_i satisfy the above equations. So the x^4 coefficient of $f(x)$ is

$$\sum \frac{1}{m_A! m_B! \dots m_F!}.$$

So the '$x^4/4!$ coefficient' is 4! times this, exactly what we want to count.

This reasoning give us that $f(x)$ as above is the exponential generating function for the a_i.

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The same reasoning gives the following.

Theorem 9.21. *Where a_i is the number of i-permutations of the multiset M containing the element i with repetition n_i for $i = 1, \ldots, d$, the exponential generating function $f(x)$ of $\{a_i\}$ is*

$$f(x) = \prod_{i=1}^{d} (1 + x + \frac{x^2}{2!} + \cdots + \frac{x^{n_i}}{n_i!}).$$

If an element i occurs infinitely many times, then we use the factor polynomial

$$e^x = 1 + x + \frac{x^2}{2!} + \ldots.$$

Example 9.22. How many ways can we colour the vertices of a n vertex (labelled) path red and blue, so that red is used at least twice.

The exponential generating function for the number of colourings is

$$f(x) = (1+x+x^2/2!+x^3/3!+\ldots)(x^2/2!+x^3/3!+\ldots) = e^x(e^x-x-1) = e^{2x}-xe^x-e^x$$

This e^{2x} expands to $1+2x+2^2x^2/2!+2^3x^3/3!+\ldots$ and xe^x to $x+x^2+x^3/2!+x^4/3!$, so the x^n coefficient is

$$2^n/n! - 1/(n-1)! - 1/n! = (1/n!)(2^n - n - 1).$$

The $x^n/n!$ coefficient is thus (2^n-n-1), which counts the number of colourings.

(Does this agree with your intuition? There are 2^n colourings, in 1 of the red is not used, and in n, it is used once.)

Problem 9.7. Show that e^{-x} the exponential generating function for $(1, -1, 1, -1, \ldots)$. What sequences do the exponential generating functions $e^x + e^{-x}$ and $e^x - e^{-x}$ generate?

Problems from the Text

Sect 9.4: 1,2,3,4,5,6,7

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10 Recurrence Relations

10.1 First Order Recurrence Relations

In this chapter we look at recursively defined sequences $(a_0, a_1, a_2, \dots,)$ of numbers. A *recurrence relation* is an equation, or rule, that defines the nth number a_n in terms of earlier numbers.

Example 10.1. The equation $a_n = 5a_{n-1}$ is a recurrence relation that is satsified by many sequences of numbers: $(0, 0, 0, \dots)$ is one. $(1, 5, 25, \dots,)$ and $(2, 10, 50, \dots,)$ are others.

Observe that upon choosing a first term a_0, (indeed any term in the example above,) the recurrence equation defined the whole sequence.

Example 10.2. Setting $a_0 = 3$ and $a_n = 5a_{n-1}$ for $n \geq 1$ defines the sequence $(a_0, a_1, a_2, \dots) = (3, 15, 75, \dots)$. One can show by induction that a_n has an explicit representation as $a_n = 3 \cdot 5^n$.

In general, the recurrence relation

$$a_0 = A, \qquad a_{n+1} = d \cdot a_n \text{ for all } n \geq 0$$

defines the sequence with explicit formula $a_n = Ad^n$ for all $n \leq 0$. Such an explicit formula is called the solution of a recurrence relation.

Problem 10.1. You invest \$1000 in the bank at .5% interest per month. How much is it worth after 2 years?

See Example 10.2 of text

The above recurrence relations are quite simple. They are called *homogeneous first order linear recurrence relations with constanst coefficients*, and we have solved them all.

A recurrence relation that can be written in the form

$$c_0 a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + d$$

is called a kth *order linear* recurrence relation. If $d = 0$, it is called *homogeneous* and if the c_i are constant, it is said to have *constant coefficients.*

In later sections we solve all homogeneous second order linear recurrences with constant coefficients, and then some non-homogeneous recurrences. These take more theory. To finish off this section we look as some special cases that we can do without much more theory.

Example 10.3. Solve the (non-homogeneous) recurrence relation defined by $a_0 = 0$ and $a_{n+1} = da_n + c$.

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Solution

Writing out the first couple terms we have that $a_1 = c$, $a_2 = c(1+d)$, $a_3 = c(1+d+d^2)$. We can show by induction that this pattern continues and has the solution $a_n = c(1+d+d^2+\cdots+d^{n-1}) = c\frac{d^n-1}{d-1}$.

Example 10.4. Solve the (non-linear) recurrence relation defined by $a_0 = 5$, and $a_{n+1}^3 = 2a_n^3$.

Solution

It isn't linear in a_n, but making a change of variable $b_n = a_n^3$, the recurrence $b_0 = 5^5$ and $b_{n+1} = 2b_n$ is linear. It has solution $b_n = 5^3 \cdot 2^n$. So $a_n = (5^3 \cdot 2^n)^{1/3} = 5 \cdot 2^{n/3}$ is the solution to our original relation.

Problems from the Text

Sect 10.1: 1,2,4,6

10.2 Second order linear homogeneous recurrence relations with constant coefficients

Second order linear homogeneous recurrence relations can be written as

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} = 0.$$

To solve this, we will guess a solution of the form $a_n = cr^n$ for non-zero c and r, and plug it in to the recurrence relation.

Now $c_0(cr^n) + c_1(cr^{n-1}) + c_2(cr^{n-2}) = 0$ if and only if $c_0 r^2 + c_1 r^1 + c_2 = 0$. This is only true if r is a root of the *characteristic equation* $c_0 x^2 + c_1 x + c_2 = 0$. Let's see how this plays out in an example.

Example 10.5. (Distinct Real Roots) Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$, $n \geq 2$, $a_0 = 1$, $a_1 = 5$.

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Solution

1) Rewriting the recurrence relation as $a_n - 5a_{n-1} + 6a_{n-2} = 0$, the characteristic equation is $x^2 - 5x + 6 = 0$.

2) The characteristic equation has *characteristic roots* $r = 2, 3$.

3) Each of these is a solution to the recurrence relation. As it is linear, any linear combination $A(2^n) + B(3^n)$ is also a solution, so we have the *general solution*

$$a_n = A(2^n) + B(3^n)$$

(for constants A and B) to the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$.

4) Using $a_0 = 1$ and $a_1 = 5$, we get a *specific solution*. From the first *initial condition* we get, $1 = a_0 = A(2^0) + B(3^0) = A + B$. From the second, we get $5 = a_1 = A(2) + B(3) = 2A + 3B$. Solving these, we get $B = 3$ and $A = -2$. So the specific solution to our recurrence relation is

$$a_n = (-2)2^n + (3)3^n = 3^{n+1} - 2^{n+1}.$$

Remark 10.6. It can be shown, using linear algebra, that the the general solution given in part 3) describes all possible solutions of the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$. Indeed, the recurrence can be rewritten $a_n - 5a_{n-1} - 6a_{n-2} = 0$. As a system of equations in the variables a_n, a_{n-1}, and a_{n-2} the solution space has dimension 2. We won't prove this, we will just believe it.

This lays out a general process for solving linear homogeneous recurrence relations with constant coefficients.

i. Find the characteristic equation.

ii. Find the characteristic roots.

iii. Find the general solution.

iv. Use the initial conditions to find the specific solution.

Try this process to solve the following.

Problem 10.2. Solve the following (shifted fibonacci) recurrence relation: $f_n = f_{n-1} + f_{n-2}$, $f_0 = 0$, and $f_1 = 1$. (See Example 10.10 of the text book. You should get that $f_n = \frac{1}{\sqrt{5}}(\frac{1+\sqrt{5}}{2})^n - \frac{1}{\sqrt{5}}(\frac{1-\sqrt{5}}{2})^n)$.

This process works for higher order relations than 2nd order, however we may have trouble finding the characteristic roots for higher order equations. We will mostly only consider 2nd order equations. The previous example was the easiest case. In general, our characteristic roots may not real. This is not a difficult complication. The more difficult complication is when they are real, but not distinct. (If they aren't real, the will be distinct.)

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Example 10.7. (Complex Roots - Necessarily Distinct) Solve the recurrence relation $a_n = 2a_{n-1} - 2a_{n-2}$ with initial conditions $a_0 = a_1 = 1$.

Solution

There characteristic equation is $x^2-2x+2 = 0$, which has roots $x = 1\pm\sqrt{-1} = 1 \pm i$. So our general solution is

$$a_n = A(1+i)^n + B(1-i)^n.$$

We now find the specific solution. For one, we have $1 = a_0 = A + B$, so $B = 1 - A$. From the second initial condition, we have

$$1 = a_1 = A(1+i) + B(1-i) = A(1+i) + (1-A)(1-i) = 2Ai + 1 - i.$$

Solving gives $A = B = 1/2$, so we get the specific solution

$$a_n = \frac{1}{2}\left((1+i)^n + (1-i)^n\right).$$

Using our knowledge of complex numbers, we observe that $(1+i)^n$ and $(1-i)^n$ are conjugate. Thus their sum is twice their real part, which is $\sqrt{2}\cos(n\pi/4)$, and so we get that $a_n = \sqrt{2}\cos(n\pi/4)$. However, it isn't assumed that we know how to deal with complex numbers, so it's fine for you to leave the solution as $a_n = \frac{1}{2}\left((1+i)^n + (1-i)^n\right)$.

Example 10.8. (Repeated real roots) Solve $a_n = 6a_{n-1} - 9a_{n-2}$, where $a_0 = 3$ and $a_1 = 10$.

Solution

The characteristic equation $x^2 - 6x + 9 = (x-3)^2 = 0$ has double root 3. In this case, we have a solution $a_n = A3^n$, for all A, but this is not all of the solutions. Observe that $n3^n$ is also a solution:

$$n3^n = 6(n-1)3^{n-1} - 9(n-2)3^{n-2}.$$

Again linearity gives that

$$a_n = A3^n + Bn3^n$$

is a solution for all A and B. This is our general soluton.

Using the initial conditions, we can sovle to get the specific solution $a_n = 3^{n+1} + n3^{n-1}$.

Problem 10.3. Solve $a_n = -2a_{n-1} - a_{n-2}$ where $a_0 = 2$ and $a_1 = 5$.

Solution

You should find that the characteristic equation has a double root -1, and get a specific solution $(2-7n)(-1)^n$.

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As long as we can find the characteristic roots, we should be able to solve higher order homogeneous linear equations with constant co-efficients.

Example 10.9. (Third Order) Solve $2a_{n+3} = a_{n+2} + 2a_{n+1} - a_n$, where $a_0 = 0, a_1 = 1$, and $a_2 = 2$.

Solution

The characteristic equation $2x^3 - x^2 - 2x + 1 = 0$ has roots $x = \pm 1, 1/2$. So the general solution is $a_n = A(1)^n + B(-1)^n + C(1/2)^n$. Now we find the particular solution. From the first initial condition, $0 = a_0 = A+B+C$, from the second $1 = a_0 = A-B+C/2$, from the third we have $2 = a_2 = A+B+C/4$. This is three equations in three unknowns, so we can solve it to get $A = 5/2$, $B = 1/6$, and $-8/3$. So $a_n = \frac{5}{2} + (\frac{-1}{6})^n - \frac{8}{3}(\frac{1}{2})^n$.

Now let's see a bit how we can use these recurrence relations.

Problem 10.4. How many binary strings (strings of '0's and '1's) of n bits have no two consecutive zeros?

Solution

Let a_n be the number of such strings. Clearly $a_0 = 1$, $a_1 = 2$, and $a_2 = 3$. An n bit string begins 1, and ends with any $n-1$ bit string, or begins with $0, 1$, and ends with any $n-2$ bit string. So $a_n = a_{n-1} + a_{n-2}$. We can solve this.

This is almost the same problem.

Problem 10.5. How many ways can we colour the vertices of a path on n vertices with four colours, so that there is no pair of adjecent blue vertices.

Solution

You should get a recurrence relation $a_n = 3a_{n-1} + 3a_{n-2}$.

Problem 10.6. How many compositions of the integer n use only the digits 1 and 2?

Problem 10.7. How many ways can you tile a $2 \times n$ checkerboard with the

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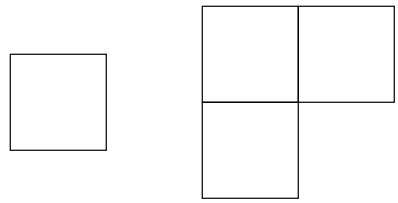

tiles of the type seen here.

This is Example 10.19 from the text.

Problems from the Text

Sect 10.2: 1,4,5,6,13,14,24,25

10.3 The Nonhomogeneous Recurrence Relation

A linear recurrence relation, written as

$$a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_d a_{n-d} = f(n)$$

is called *nonhomogeneous* if $f(n)$ is not identically 0.

We will look at the first and second order cases. Again, one can show by linear algebra that to find the general solution to such a relation, one only needs to find the general solution $a_n^{(h)}$ to a *corresponding homogeneous recurrence relation*, and then shift it by one *particular solution* $a_n^{(p)}$.

We start with an example.

Example 10.10. Solve $a_n = 2a_{n-1} + 7 \cdot 5^n$ for $n \geq 1$ and $a_0 = 4$.

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Solution

First we find the general solution to the corresponding homogeneous recurrence $a_n - 2a_{n-1} = 0$. This has characteristic equation $x - 2 = 0$ which has root $x = 2$ so we have the general soluton $a_n^{(h)} = A \cdot 2^n$.

Now we want a particular solution to $a_n - 2a_{n-1} = 7 \cdot 5^n$. Lets guess that there is one of the form $c \cdot 5^n$. (We talk about why after this example.) Plugging into the recurrence relation, we get

$$c \cdot 5^n - 2c \cdot 5^{n-1} = 7 \cdot 5^n,$$

which is true if and only if $c \cdot 5 - 2c = 35$, or $c = 35/3$. So $a_n^{(p)} = (35/3) \cdot 5^n$ is a solution to the recurrence.

The general solution to the recurrence is thus

$$a_n = a_n^{(h)} + a_n^{(p)} = A \cdot 2^n + (35/3) \cdot 5^n.$$

Using the initial conditions we now find the specific solution. We have $4 = a_0 = A(2^0) + (35/3)5^0 = A + 35/3$, which gives $A = -23/3$. The specific solution is thus

$$a_n = (35/3)5^n - (23/3)2^n = (1/3)(35 \cdot 5^n - 23 \cdot 2^n).$$

For $f(n) = 7 \cdot 5^n$ we guessed a solution of the form $c \cdot 5^n$. The solution that we guess is suggested by the form of $f(n)$:

When $f(n)$ is	$a_n^{(p)}$ is of the form
c	A
cn	$An + B$
cn^2	$An^2 + Bn + C$
cr^n	Ar^n
cn^2r^n	$r^n(An^2 + Bn + C)$

From the above, you can interpolate appropriate guesses for various other special cases of $f(n)$. You will be responsible only for solving nonhomogeneous equations with $f(n)$ from the above table.

Problem 10.8. Solve $a_n = -6a_{n-1} - 9a_{n-2} + (-3)^n$ for $n \geq 2$, where $a_0 = 2$ and $a_1 = 3$.

Problem 10.9. (The Towers of Hanoi) In the game of the Towers of Hanoi one has three pegs. A stack of disks of different radii are stacked on one peg with no disk on top of a larger disk. The game is to move the stack of disks to another peg. In a move, one can move one disk, placing it on an empty peg, or on a peg of smaller radius. How many moves can this be done in when there are n disks.

See Example 10.28 of text.

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Problems from the Text

Sect 10.3: 1,6,7,8

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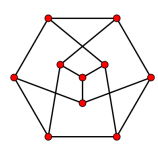
11 More Graph Theory

Much of This is from Chapter 11 - 12 of the text, some of it is not.

Graph theory studies the properties that arise in a structure due to the simple idea of a relationship between elements of a set. Many such properties are purely combinatorial, others have more structural ramifications, which arise surpisingly in most any field of mathematics.

Here are some typical questions that we might ask of this, or any graph. The definitions you need to understand these questions come later, but they are probably what you think they are.

- How many colours do we need, when colouring the vertices of P to be sure that adjacent vertices get different colours.
- Can you draw P without intersecting edges?
- How many edges (vertices) must you remove to *disconnect* P?
- For any two pairs of vertices u, u' and v, v' can you find a *path* between u and u', and another between v and v', such that the two are disjoint?
- What is the smallest *cycle* in P?
- How many 5-cycles are there in P?
- Can we partition the edges of P into disjoint 5-cycles?
- Can we find a *walk* in P that visits every vertex (edge) exactly once?
- Is P the same graph as this: 

11.1 Homomorphisms

Definition 11.1. A *homomorphism* $f : G \to H$ of a graph G to another graph H, is a vertex map $f : V(G) \to V(H)$ such that

$$uv \in E(G) \Rightarrow f(u)f(v) \in E(H).$$

If there is a homomorphism $f : G \to H$, we write simply $G \to H$.

Any isomorphism is a homomorphism, and as usual, a homomorphism $f : G \to H$ is an isomorphism if and only if there is an inverse homomorphism

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$f^{-1} : H \to G$ such that $f \circ f^{-1}$ and $f^{-1} \circ f$ are the identity maps. But there is another property you may be used to for other finite structures, that doesn't hold for us: **a bijective homomorphism is not necessarily an isomorphism.** Indeed, there are many bijections that are not isomorphisms.

Example 11.2. For any graph G on n vertices there is a bijective homomorphism $f : G \to K_n$.

A homomorphism $f : G \to H$ is a map on the vertices, but it induces a map $f_E : E(G) \to E(H)$ on the edges:

$$f_E(uv) = f(u)f(v).$$

The homomorphism f is called *edge-injective* if f_E is injective, and *edge-surjective* if f_E is surjective.

Problem 11.1. Show that if f is injective, then it is edge-injective, but that the converse is not true.

Problem 11.2. Show that f is an isomorphism if and only if it is bijective and edge-surjective.

Problem 11.3. Show that $C_{2n+1} \to C_{2m+1}$ if and only if $m \leq n$. Show that $C_n \to K_2$ if and only if n is even. Show that $G \to K_2 \Rightarrow C_{2n+1} \not\to G$ for every n.

Problem 11.4. Show that a bijective homomorphism $f : G \to H$ is an isomorphism if and only if it is also a homomorphism of $\overline{G} \to \overline{H}$.

11.2 Degrees and Degree Sequences

Definition 11.3. For a vertex v in a graph G, the *degree* of v, written $\deg(v)$, is the number of edges in G incident to v. The *degree sequence* of a graph G is the multiset of the degrees of the vertices of G. The minimum degree, $\delta(G)$, and maximum degree, $\Delta(G)$, of a graph G are the minimum and maximum degrees respectively in its degree sequence.

Problem 11.5. Show that every graph contains a path of length at least $\delta(G)$.

Clearly, degree sequence, minimum degree, and maximum degrees are graph properties. So the following two graphs, which have the same number of vertices, and edges are not isomorphic as they have different degree sequences: $(4, 2, 2, 2, 2)$ and $(3, 3, 2, 2, 2)$ respectively, (and in fact different maximum degrees.)

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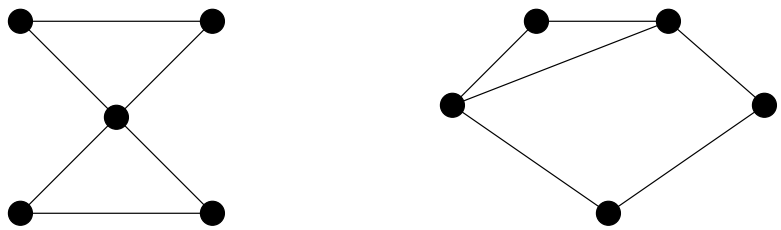


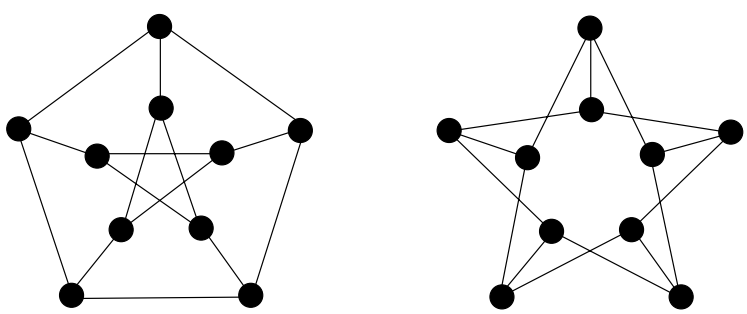
Here is a simple observation about degree sequence:

Theorem 11.4. *If $(d_1, d_2, \ldots, d_n)$ is the degree sequence of a graph, then the degree sum: $\sum_{i=1}^{n} d_i$ is equal to $2|E(G)|$.*

Proof. Each edge of the graph contributes 2 to the degree sum, one for each endpoint. □

Corollary 11.5. *In any graph G, there is an even number of vertices of odd degree.*

Problem 11.6. What are the degree sequences of the following graphs? Are the following two graphs, with the same degree sequences, isomorphic?



Problem 11.7. What is the degree sequence of the cube Q_n?

Definition 11.6. A graph G is called *k-regular* if every vertex has degree k.

Problem 11.8. How many edges does a k-regular graph on n vertices have?

A sequence $(d_1, d_2, \ldots, d_n)$ non-increasing integers is *graphic* if it is the degree sequence of a graph. We saw above that one necessary condition for a

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sequence to be graphic is that it sums to an even number. But this is not sufficient: $(4, 2, 0)$ is not graphic.

Problem 11.9. Show that the following condition is also necessary for a sequence $(d_1, d_2, \ldots, d_n)$ to be graphic. For all $1 \leq k \leq n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(k, d_i).$$

In fact, this condition is also sufficient. But this is fair bit harder to show:

Theorem 11.7. *(Erdős-Gallai '60) A sequence $(d_1, d_2, \ldots, d_n)$ is graphic if and only if $\sum_{i=1}^{k} d_i$ is even and for all $1 \leq k \leq n$,*

$$\sum_{i=1}^{n} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(k, d_i).$$

The Erdős-Gallai condition is what we use to decide if a sequence is graphic, but there is also a recursive method to decide it. It depends on the following idea:

Lemma 11.8. *(Havel-Hakimi) If a sequence $D = (d_1, d_2, \ldots, d_n)$ is graphic, then there is a graph with degree seqence D such that the maximum degree vertex is adjacent to the d_1 vertices of next highest degree.*

Thus to decide if a sequence D, such as $(3, 3, 2, 2, 2)$, is graphic, the *Havel-Hakimi algorithm* is as follows. While D is not all zeros,

- remove d_1, and
- remove 1 from the next d_1 degrees $d_2, \ldots, d_{d_1-1}$,

if possible.

The algorithm clearly terminates. If it terminates with an empty sequence, (or a sequence of zeros), then D is graphic. Otherwise, it is not graphic.

Starting with $(3, 3, 2, 2, 2)$, we get $(\cancel{3}, 3-1, 2-1, 2-1, 2) = (2, 2, 1, 1)$, and then $(1, 0, 1) = (1, 1)$, and then $(0, 0)$. So $(3, 3, 2, 2, 2)$ is graphic.

After one definition, let's prove the Lemma.

Definition 11.9. The *neighbourhood* $N(v)$ of a vertex v in a graph G is the set

$$\{u \in V(G) \mid uv \in E(G)\}$$

of all neighbours of v.

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Proof. Let $D = (d_1, d_2, \ldots, d_n)$ be a graphic sequence. Given a realisation G of D, let v be a vertex of degree $\Delta = d_1$, and let $T = (v_2, \ldots, v_{\Delta+1})$ be a set of Δ vertices where v_i has degree d_i for all i. If $N(v) = T$ then we are done, so towards contradiction, assume that the realisation G of D that maximises $\Delta' = |N(v) \cap T|$ has $\Delta' < \Delta$, and assume that G is such a realisation.

We find a new realisation G' with $|N(v) \cap T| > \Delta'$, so the lemma follows by contradiction.

As $N(v) \neq T$, there is $t \in T$ with $v \not\sim t$ and $s \notin T$ with $v \sim s$. Furthermore, since $t \in T$ and $s \notin T \cup \{v\}$, $\deg(t) \geq \deg(s)$. So there is some vertex w with $w \sim t$ and $w \not\sim s$.

Removing the edges vs and tw from G and adding vt and sw we get a graph G' with the same degrees on every vertex, (so another realisation of D), but with $|N_{G'}(v) \cap T| = |(N_G(v) \cap T) \cup \{t\}| = \Delta' + 1$, as needed for the contradiction. $\square$

Problem 11.10. Decide, using Havel-Hakimi, whether or not the sequence $(5, 5, 5, 3, 2, 2, 1, 1)$ is graphic.

11.3 Walks, trails, paths, circuits and cycles

Definition 11.10. Let x and y be vertices in a graph G. An xy-walk is a finite alternating sequence of vertices and edges:

$$x = x_0, e_1, x_1, e_2, e_2, \ldots, x_{n-1}, e_n, x_n = y$$

where $e_i = x_{i-1}x_i$ for $i = 1, \ldots n$.

Usually we just write the vertices

$$x_0, x_1, \ldots, x_n,$$

or, if x and y are known, the edges

$$e_1, e_2, \ldots, e_n.$$

A walk in a graph G can be viewed as a homomorphism of a path P_n to G. The *length* of the walk is n, the number of edges in it. If a walk is edge-injective, it is called a *trail*; if it is injective, it is called a *path*.

An xy-walk is *closed*, or a *circuit* if $x = y$. Though a circuit may be denoted $v_1, \ldots, v_m$, we usually write $v_1, \ldots, v_m, v_1$ to emphasise the fact that it is closed. A circuit can be viewed as a homorphism of a cycle C_n to G. If it is injective, it is called a *cycle*. The *girth* of the cycle is n.

Theorem 11.11. *Any xy-trail in a graph has an xy-path as a subgraph.*

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Intuitively, we just start with an xy-trail and 'chop' out the repeated bits until it is a path. A lot of times it is very hard to write the intuitive proof rigorously. Try it before you proceed. A nice way to prove this theorem is the following proof by contradiction.

Proof. By assumption, there exists an xy trail in G. Let $x = x_0, x_1, \ldots x_d = y$ be the shortest xy trail.

Towards contradiction, assume that the trail isn't a path. Then there exist some $0 \leq i < j \leq d$ such that $x_i = x_j$. But then

$$x_0, x_1, \ldots, x_{i-1}, x_i, x_{j+1}, \ldots, x_d$$

is an xy trail that is shorter than the original one. This is a contradiction, so this path is a trail. □

Problem 11.11. Prove the Theorem 11.11 using induction.

Here is a similar problem. We prove it by contradiction using what is called a minimum counterexample.

Problem 11.12. Prove the following. Any odd circuit in a graph has an odd cycle as a subgraph.

Proof. Assume that the statement is false. Then there is a counter-example to the statement; that is, there is a graph G containing an odd circuit $C = v_1, \ldots, v_m, v_1$ which contains no odd cycle as a subgraph. We may assume that this is a counter-example that minimizes the size m of the circuit C.

Now, as C contains no odd cycle, then, in particular, it is not an odd cycle. So there exist i, j with $1 \leq i < j \leq m$, such that $v_i = v_j$. But then both $C_1 = v_i, \ldots, v_{j-1}, v_j = v_i$ and $C_2 = v_j, v_{j+1}, \ldots, v_m, v_1, v_2, \ldots, v_i$ are smaller circuits in G which are subgraphs of C, so can also contain no odd cycles. However, one of them is odd, so is a smaller counter-example to the statement than C. This contradicts our choice of C. □

Definition 11.12. The *girth* of a graph is the minimum girth of all cycles in the graph.

Problem 11.13. What is the girth of the cube Q_n? ..of the Peteresen graph?

Problem 11.14. Show that for any vertex u and any other vertex v of the Petersen graph P, there is an *automorphism* f (that is, an isomorphism $f : P \to P$) such that $f(u) = v$. Graphs with this property are called *vertex transitive.*

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Distance

Definition 11.13. The *distance*, $\mathrm{dist}(u, v)$ between two vertices u and v in a graph G is the length of the shortest uv-walk in G. The *diameter* $\mathrm{diam}(G)$ of G is the maximum $\mathrm{dist}(u, v)$ over all pairs u and v of vertices in G. The *radius* $\mathrm{rad}(G)$ of G is

$$\min_{u \in V(G)} \max_{v \in V(G)} \mathrm{dist}(u, v).$$

A vertex u that minimises $\max_{v \in V(G)} \mathrm{dist}(u, v)$ is called central.

Problem 11.15. What is the diameter of K_n, C_n, Q_n?

Problem 11.16. Show that for a graph G containing a cycle, the girth of G is at most $2\mathrm{diam}(G) + 1$.

Problem 11.17. How many vertices can a graph of radius k and maximum degree Δ have?

Connected Graphs

Definition 11.14. A graph G is called *connected* if there is an xy-path for every two vertices $x, y \in V(G)$.

Any *maximal* connected subgraph of G, that is, a connected subgraph that is not a proper subgraph of any other subgraph of G, is called a component of G.

Note

The word 'maximal' is different from 'maximum' and is not really used outside of mathematics. The word maximal is defined with respect to a *partial ordering*. (See Section 7.3 of the text for more about partial orders.) An element in a partial ordered set is *maximal* if it is greater than all elements with which it is comparible.
For graphs there are several different partial orders that we might talk about consider.

Generally, when we say 'maximal' we are talking about the partial order 'subset inclusion' a graph is maximal in a family of graphs if it is not a subgraph of any graph in the family but itself.

Example 11.15. Let G be the graph containing a K_4 on the vertices $\{1, 2, 3, 4\}$ and a P_2 on the vertices $\{a, b, c\}$. G has two components, K_4 and P_2.

Eulerian Graphs

It is often told that the first problem of graph theory was the problem of the Bridges of Königsberg. In the fictional city of Königsberg (Problem: prove or

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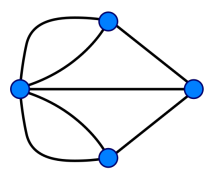
disprove my unpopular claim that Königsberg is fictional.) there were seven bridges. The dandies of the city, on a slow Sunday, would walk about the town, playing a riddlish game. Starting whereever they pleased, they were to walk around, crossing every bridge, without crossing any bridge more then once. This game went on for centuries, and several liars claimed to have done it. Fermat once wrote in the margin of a book that it was simple enough. Until eventually, Euler proved it was impossible, and stabbed Fermat in a dual. (Prove that any of this is true.)

The city of Königsburg consisted of four dots, connected by seven bridges. It looked a lot like this:



How Euler proved it is lost in the depths of historical fiction. But our story inspires the following definitions.

Definition 11.16. An trail in G that contains all edges of G is called an *euler trail*; a closed euler trail is called an *euler circuit*. A graph is called *eulerian* if it contains an euler circuit.

Viewing a walk in G as a homomorphism from a path, an euler trail can be viewed as an edge-bijective walk in G. Similarily an euler circuit can be viewed as an edge-bijective circuit in G.

It is clear that a graph that is disconnected, with edges in each component, cannot be eulerian.

The following is what Euler claims to have proved.

Theorem 11.17. *A connected graph is eulerian if and only if all of its vertices have even degree.*

Before we prove the theorem, we prove a lemma which does the easy direction for us.

Lemma 11.18. *Let $T = \{v_1, \ldots, v_m\}$ be a trail in G. The every vertex in T, except possibly v_1 and v_m have even degree (in T). Further, v_1 and v_m have even degree (in T), if and only if T is closed (ie, if and only if $v_1 = v_n$).*

Proof. For each occurence of a vertex v in the multiset $\{v_2, \ldots, v_{m-1}\}$, v is in exactly two edges in T. Further, for each occurence in the multiset $\{v_1, v_m\}$, it is in exactly one edge in T. As T is a trail, each edge is distinct, so the lemma follows. $\square$

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Now we prove the theorem.

Proof. If G is an eulerian graph, we have already observed that it must be connected. The lemma tells us that each vertex must have even degree; indeed, in an eulerian trail T of G, the edges of T are exactly those of G.

Thus letting G be a conected graph in which every vertex has even degree. We must show that it is eulerian. That there is a trail in G is clear, so let $T = v_1, v_2, \ldots, v_m$ be a longest trail in G.

First observe that T must be closed. Indeed, if it isn't, then the degree of v_m in G is odd by the lemma, so there is another edge in G incident to v_m. Adding this to T, we would get a longer trail,contradicting the choice of T. So T is a circuit.

If T does not contain every edge, then $G' = G \setminus T$ is non-empty. As G is connected, some edge $e = u_1u_2 \in G'$ has at least one endpoint $u_1 = v_i$ in $V(T)$.

But then

$$u_2, u_1(= v_1), v_2 \ldots, v_m$$

is a longer trail in G than T, contradicting the choice of T. So T contains every edge, and so G is eulerian. □

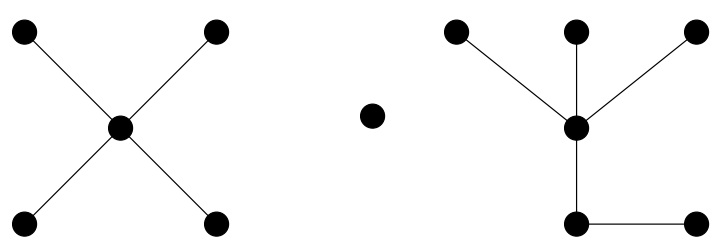
Problem 11.18. Give necessary and sufficient conditions for a graph to have an euler trail. Does Königsberg satisfy these conditions?

Problem 11.19. Does the cube Q_n have an euler trail? An euler circuit?

11.4 Trees

Definition 11.19. A *forest* is a graph with no cycles as subgraphs. A *tree* is a connected forest.

The following graph is a forest, each of its three components is a tree.



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Definition 11.20. A *spanning tree* of a graph G is a spanning subgraph which is a tree.

Theorem 11.21. *A graph G is connected if and only if it has a spanning tree.*

Proof. Let G be a graph with a spanning tree T. Let x and y be any two vertices in G. As T is spanning, they are also in T. As T is connected, there is an xy-path $P = e_1e_2 \dots e_d$ in T. But $E(T) \subset E(G)$ so P is in G as well, that is, it is an xy-path in G. This was for any $x, y \in V(G)$, so G is connected.

On the other hand assume that G is connected. Let H be a minimal connected spanning subgraph of G, that is, a connected spanning subgraph, which itself has no connected spanning proper subgraph. We show by contradiction that H is a tree. Assume that it isn't. Then it contains a cycle $v_1v_2 \dots v_d$. Let $H' = H \setminus v_1v_2$. If H' is disconnected, then there exist vertices x and y of H such that there are no xy-paths in H'. So all xy-paths in H use v_1v_2, that is, they are of the form $x = x_0x_1 \dots x_iv_1v_2x_{i+1} \dots x_m = y$. But then $x_0x_1 \dots x_iv_1v_nv_{n-1} \dots v_2x_{i+1} \dots x_m$ is an xy-walk in H', contradicting the fact that there is no xy-path in H'. Thus H' is a smaller connected spanning subgraph, contradicting the choice of H. So by contradiction H is a tree. □

Definition 11.22. A *leaf* or *pendant edge* of a tree is a vertex of degree one.

Lemma 11.23. *Any tree has a leaf.*

Proof. Let G be a graph with no leaves. We show that G has a cycle, so is not a tree. This gives the contrapositive of the Theorem. Let $P = v_1v_2 \dots v_n$ be a longest path in G. As v_1 has some neighbour other than v_2 and there is no longer path than P, v_1 has a neighbour v_d on P, so $v_1v_2, \dots v_dv_1$ is a cycle in G. □

Theorem 11.24. *For a tree T, $|V(T)| = |E(T)| + 1$.*

Proof. The proof is by induction on $|V(T)|$. Since the only tree on one vertex is K_1 and the theorem is true for K_1 we assume that $n \geq 2$, and that the theorem is true for graphs on less than n vertices.

As T is a tree, it has a leaf v. Let $T' = T \setminus v$, the graph we get from T by removing v.

Claim 11.25. *T' is a tree*

Proof of claim. To see that it is connected, observe that the only paths in T that contain v, contain v as an endpoint. So for any vertices $x, y \in V(T) \setminus \{v\}$, the xy-path connecting them in T is also in T'. As $T' \leq T$ it has no cycles. So T' is a tree. ◇

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Now $|V(T)| - 1 = |V(T')|$, which by induction is $|E(T')| + 1 = |E(T)|$. So $|V(T)| = |E(T)| + 1$ as needed. $\square$

Definition 11.26. A *cut edge* of a graph G is any edge whose removal increases the number of components of G.

Theorem 11.27. *Let $G = (V, E)$ be a graph. Then the following are equivalent.*

i. G is a tree.

ii. G is connected, but every edge is a cut edge.

iii. G has no cycles, and $|V| = |E| + 1$.

iv. G is connected, and $|V| = |E| + 1$.

v. Any two vertices of G are connected by a unique path.

Proof. From the previous theorem we have that (i) implies (iii) and (iv). We show that (iii) $\Rightarrow$ (i) and (iv) $\Rightarrow$ (ii) $\Rightarrow$ (i). This shows the equivalence of the first four statements. As an exercise, show that statement (v) is also equivalent.

(iii) $\Rightarrow$ (i): Assume that G has no cycles, so each component of G is a tree. For each component C we thus have that $|V(C)| = |E(C)| + 1$. So $|V(G)| = |V(E)| + c$ where c is the number of components. By $|V| = |E| + 1$, there is only one component, so it is a tree.

(iv) $\Rightarrow$ (ii): Assume that G is connected an has $|V| = |E| + 1$. If there is some edge e that is not a cut-edge, then $G' = G \setminus \{e\}$, the graph we get by removing e, is connected, and so contains a spanning tree T. Thus $|V| = |V(G')| = |V(T)| = |E(T)| + 1 \leq |E(G')| + 1 = |E| + 2$, but this is a contradiction.

(ii) $\Rightarrow$ (i): Let G be connected and every edge of G be a cut-edge. As G is connected it has a spanning tree. Since every edge is a cut edge, no proper subgraph has a spanning tree, so G itself is a spanning tree.

$\square$

Problems from the Text

Sect 12.1: 1a,3,7,8,13

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11.5 Bipartite and Planar Graphs

Bipartite graphs

Definition 11.28. A graph G is *bipartite* if and only if there is a partition $A \cup B = V(G)$ of its vertices (so $A \cap B = \emptyset$), such that all edges are between A and B, (ie, that A and B both induce empty sets.

Problem 11.20. Show that if G is bipartite if and only if $G \to K_2$. Show that $C_{2n+1} \to G$ implies that G is not bipartite.

We can say more

Proposition 11.29. *For a graph G, the following are equivalent*

i. *G is bipartite.*

ii. $G \to K_2$.

iii. *G contains no odd cycles.*

iv. $C_{2n+1} \not\to G$ *for all integers* $n \geq 1$

Proof. We've proved most of this over the course of these notes. With impliction iii) $\Rightarrow$ ii), the proof can be pieced together from the exercises. So we prove this implication.

Assume that G has no odd cycles. We define a homomorphism $f : G \to K_2$ as follows. Let v_0 be some vertex in G. For every vertex v in G let

$$f(v) = \begin{cases} 0 \text{ if } \operatorname{dist}(v, v_0) \text{ is even} \\ 1 \text{ if } \operatorname{dist}(v, v_0) \text{ is odd} \end{cases}$$

We claim that this is a homomorphism. If it is not, there are some two vertices adjacent u and v such that $f(u) = f(v)$. That is, $\operatorname{dist}(u, v_0)$ and $\operatorname{dist}(v, v_0)$ have the same parity. But then there is path P_1 from v_0 to v and P_2 from u to v_0 of the same parity, so the path $P_1 + vu + P_2$ is an odd path from v_0 to v_0. That is, it is an odd circuit in G. But we showed this implies there is an odd cycle in G, which contradicts our assumption. $\square$

Problem 11.21. Let G be a graph with n vertices and $m > (n/2)^2$ edges. Show that G cannot be bipartite.

Planar Graphs

Can we draw the petersen graph in the plane so that no edges cross?

A graph G is *planar* if it can be drawn in the plane with no edges crossing. A drawing of the graph in the plane with no edges crossing is called an *(planar) embedding* of G.

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Problem 11.22. Show that K_4 is planar but that K_5 is not.

We can show that K_5 is not planar by ad-hoc arguments, but such arguments difficult for larger graphs. That's okay though. That K_5 is not planar tells us that a lot of other graphs are not planar.

Problem 11.23. Show that no graph containing K_5 as a subgraph, can be planar. Show that the petersen graph is not planar.

Definition 11.30. A *subdivision* of a graph G is a graph we get from G by dividing edges. That is, G' is a subdivision of G if we can construct it by repeated application of the following alteration, called an *edge subdivision.*

- remove an edge $e = ab$,
- add a new vertex v_e,
- add the edges av_e and bv_e.

It should be pretty clear that because K_5 is not planar, no subdivision of K_5 is planar. So no graph containing a subdivision of K_5 as a subgraph is planar. Can you think of any other graphs that are not planar? Please don't look ahead. It will ruin the surprise.

Okay. Here's the surprise. $K_{3,3}$ is also not planar, and has no subdivision of K_5. But that is essentially it. Kuratowski showed the following.

Theorem 11.31 (Kuratowski's Theorem)**.** *A graph is planar if and only if it contains no subdivision of K_5 or $K_{3,3}$ as a subgraph.*

We won't prove this, as it is a little involved. But we should show the following.

Problem 11.24. Show that $K_{3,3}$ is not planar.

One can do this with ad-hoc arguments. But we can also do it more methodically.

Euler's Formula

A planar embedding of a graph G defines *faces* in the plane: the little bits of plane you get by cutting along all the edges of the embedded graph. Where $|V(G)| = n$ and $|E(G)| = m$ and f is the number of faces. Euler showed the following fundamental identity.

Theorem 11.32 (Euler's Formula)**.** *For any embedding of a connected graph G in the plane,*

$$n - m + f = 2.$$

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Proof. The proof is by induction on m. As G is connected, $m \geq n-1$. If $m = n-1$ then G is a tree, so there is only one face, and the identity holds. This is our base case.

Now assume that $m \geq n$. Then there is an non-cut edge e. Remove it. Its removal joins two faces, so reduces f by one, but also reduces e by one. So the identity holds from the induction hypothesis. □

Corollary 11.33. *If G is connected and planar and $n \geq 3$, then*

$$m \leq 3n - 6.$$

Proof. Count the number N of pairs (e, r) where e is an edge, and r is a face with e on the boundary. As every edge is in at most 2 faces, we have that $N < 2m$ and as every face has at least 3 edges, $3f < N$ so

$$f \leq N/3 \leq \frac{2}{3}m.$$

Plugging this into $n - m + f = 2$ gives that

$$n - m/3 = n - m + \frac{2}{3}m \geq 2$$

so $m \leq 3n - 6$ as needed. □

Problem 11.25. Use this corollary to show that K_5 and $K_{3,3}$ are not planar.

Problem 11.26. ?? Show that any planar graph has a vertex of degree at most 5.

Dual Graphs

Definition 11.34. Given an embedding of a graph G we construct a new (multi-)graph G^*, called the *dual of G,* from G as follows.

The vertices of G^* are

$$V(G^*) = \{v_f \mid f \text{ is a face of } G\ \}.$$

The edges are defined as follows. For every edge e of G let $e^* = v_f v_g$, where f and g are the faces of G incident to e. Then $E(G^*)$ is the **multiset**

$$E(G^*) = \{e^* \mid e \in E(G)\}.$$

(Note that both $V(G^*)$ and $E(G^*)$ depend on the particular embedding we chose for G.)

Not only is G^* a multigraph, it can have loops:

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Problem 11.27. What is the dual of a tree on n vertices.

Problem 11.28. Find conditions on a graph G so that its dual G^* has no loops. Find conditions on G so that its dual, under any embedding, has no multiedges.

Problem 11.29. Give an embedding of K_4 and construct its dual. Can you find an embedding of K_4 that gives a different dual? Can you find some graph G with two different embeddings that yield different duals?

Problem 11.30. Show that the dual of an embedded graph is planar.

Theorem 11.35. *Let G be planar and G^* the dual of an embedding of G. Then G is bipartite if and only if G^* is eulerian.*

Proof. One way is easy. Let G be a bipartite graph embedded in the plane. For any vertex v_f of G^*, the walk around the boundary of f is a closed walk. As G is bipartite, it is even, so v_f has even degree. This was for any vertex of G^*, so G^* is euleriain.

On the other hand, assume that G^* is eulerian. We want to show that every cycle of G is even. (This is clear of the boundary of faces under a given embedding but not of every cycle.) Fix an embedding of G.

Let C be a cycle in G, and let $A \subset F(G)$ be the set of faces inside C relative to the embedding. Then C is the symmetric difference of the edge sets of faces in A. Each face has a even number of edges, so so does their symmetric difference. Thus C is even. □

Problem 11.31. Draw an embedding of Q_3. Draw its dual.

11.6 Hamilton Paths and Cycles

Definition 11.36. A path, or cycle, in G is *hamilton* if it is bijective. A graph G is *hamiltonian* if it contains a hamiltonian cycle.

There were some nice conditions that characterised the eulerian graphs. We are not so lucky with hamilton graphs. In fact, the problem of deciding if a graph on n vertices is hamiltonian, is NP-complete. This means the best known algorithm we have for deciding, takes time that is exponential in in n, (at least for some graphs). This doesn't mean we can't say anything about graphs that are hamiltonian. Indeed they must be connected, and have no leaves. We can say a bit more.

Problem 11.32. Find a hamilton path in Q_n.

Problem 11.33. How many hamilton cycles are there in the n-wheel W_n. The W_n is the n-cycle C_n with one more vertex h adjacent to all other vertices.

Here is an early result of Dirac.

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Theorem 11.37 (Dirac 1952)**.** *A graph G with n vertices is hamiltonian if $\delta(G) \geq n/2$.*

Instead of proving this we prove a slightly stronger result of Ore.

Theorem 11.38 (Ore 1960)**.** *A graph G with $n \geq 3$ vertices if hamiltonian if for every two vertices $x \neq y$ with $x \not\sim y$, we have* $\deg(x) + \deg(y) \geq n$.

Proof. Towards contradiction, let G be a maximal counterexample on $n \geq 3$ vertices. So G is not hamiltonian, and satisfies

$$(*) \qquad x \not\sim y \Rightarrow \deg(x) + \deg(y) \geq n,$$

but $G + xy$ is hamiltonian for any $x \not\sim y$. (We may assume that G is not complete, as complete graphs are clearly hamiltonian.) Let $H = G + xy$ for some $x \not\sim y$ in G. Then H is hamiltonian, and any Hamilton cycle contains xy. So there is a hamilton path

$$x = x_1 \sim x_2 \sim \ldots x_n = y.$$

Now if $x_1 \sim x_i$ and $x_{i-1} \sim x_n$, then

$$x_1 \sim x_2 \sim \cdots \sim x_{i-1} \sim x_n \sim x_{n-1} \sim \cdots \sim x_i \sim x_1$$

is a hamilton cycle not using xy so in G, which is a contraction. So for each of x_1s $\deg(x_1)$ neighbours x_i in $\{x_2, \ldots, x_{n-1}\}$ (recall that $x_1 \not\sim x_n$), x_{i-1} is not a neighbour of x_n. So

$$\deg(x) + \deg(y) = \deg(x_1) + \deg(x_n) \leq \deg(x_1) + (n - 1 - \deg(x_1)) \leq n - 1$$

This contradicts $(*)$, so the theorem is true.

□

Problem 11.34. Prove Diracs theorem using Ore's theorem.

Problem 11.35. Do problems 10, 18, 21 from Section 11.5 of the text.

11.7 Graph Colouring

Definition 11.39. A *(proper) k-colouring* of a graph G is a function $f : V(G) \to [k]$ such that

$$u \sim v \Rightarrow f(u) \neq f(u).$$

If G has a k-colouring, it is *k-colourable*. The *chromatic number* $\chi(G)$ of G is the minimum k for which G is k-colourable. If $k = \chi(G)$, we also say that G is *k-chromatic*.

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Problem 11.36. For what graph H is the following statement true? A graph G is k colourable if and only if $G \to H$.

Problem 11.37. What is the chromatic number of:

i. K_n

ii. C_n

iii. Q_n

iv. W_n

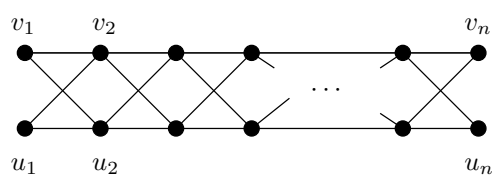
v. The petersen graph.

Problem 11.38. Describe all 2-colourable graphs. (That is, complete the statement " A graph is 2-colourable if and only if ... " using properties we've seen before.)

Problem 11.39. (This is hard.) Let $G(n)$ be the graph (pictured below) with vertices $V(G) = \{v_1, \ldots, v_n\} \cup \{u_1, \ldots, u_n\}$ and the following edges.

- $v_i \sim v_j$ if $|i - j| = 1$.
- $u_i \sim v_j$ if $|i - j| = 1$.
- $u_i \sim u_j$ if $|i - i| = 1$.

How many 3-colourings does the following graph G have? (A recursive answer is fine.)



Definition 11.40. The *clique number* $\alpha(G)$ of a graph G is maximum n such that K_n is a subgraph of G.

Problem 11.40. Show that $\chi(G) \geq |V(G)|/\alpha(\overline{G})$.

Definition 11.41. A graph G is k-critical if it is k-chromatic, but any proper subgraph is $(k-1)$-colourable.

Problem 11.41. Show that K_k is k-critical.

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Problem 11.42. Show that any k-chromatic graph contains a k-critical graph as a subgraph.

Problem 11.43. Find all 3-critical graphs.

Problem 11.44. Show that any n-critical graph G has $\delta(G) \geq n - 1$.

From the last two problems and Problem **??** you should be able to do the following.

Problem 11.45. Show that any planar graph is 6-colourable.

In fact, more is true:

Theorem 11.42. *(The Four-Colour Theorem [Haken-Appel 1976]) Any planar graph is* 4*-colourable.*

The proof of this is very long and difficuly. But it is an interesting exercise (maybe also a bit difficult) to prove that every planar graph is 5-colourable.

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